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Estimating Remaining Useful Life and State of Health of EV Batteries by Hybrid Physics-Informed Deep Learning Framework

A Thesis

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THESIS CERTIFICATE

This is to undertake that the Thesis titled **ESTIMATING REMAINING USEFUL LIFE AND STATE OF HEALTH OF EV BATTERIES BY HYBRID PHYSICS-INFORMED DEEP LEARNING FRAMEWORK**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master Of Technology in Industrial Artificial Intelligence**, is a bona fide record of the research work done by me under the supervision of **Dr. Sandeep Patalay**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS State of Health (SOH); Remaining Useful Life (RUL); Physics-Informed Learning; Hybrid Models; Gated Recurrent Unit (GRU); Battery Degradation; Lithium-ion Batteries; Prognostics and Health Management (PHM); Edge-compatible architecture; ODE Regularization; NASA PCoE; CALCE

The accelerated electrification of the global automotive sector, driven by the imperative to decarbonize transportation and meet international climate accords, has positioned the Lithium-ion (Li-ion) battery as a critical asset in modern energy infrastructure. However, the widespread adoption of Electric Vehicles (EVs) is currently hindered by the “range anxiety” phenomenon and the economic uncertainty surrounding battery longevity and resale value. Accurate estimation of the State of Health (SOH) and prediction of the Remaining Useful Life (RUL) are paramount to ensuring the safety, reliability, and economic viability of these electrochemical systems. The domain of Prognostics and Health Management (PHM) currently faces a fundamental dichotomy between model-based approaches, which offer interpretability, but suffer from high computational complexity and parameterization difficulties, and data-driven methods, which provide high accuracy, but lack physical explainability and generalizability.

This report details the foundational research for a master’s thesis aimed at bridging this gap through a Hybrid Physics-Informed Deep Learning Framework. Unlike traditional Physics-Informed Neural Networks (PINNs) that attempt to solve computationally expensive Partial Differential Equations (PDEs) for ion diffusion and electrochemical kinetics, this research proposes a “Physics-Regularized” Gated Recurrent Unit (GRU) network. This novel architecture embeds tractable empirical Ordinary Differential Equations (ODEs)—specifically the Verhulst logistic model and the Double-Exponential model—directly into the loss function of the deep learning model. By utilizing a

composite loss function weighted by homoscedastic uncertainty, the framework aims to constrain the neural network’s optimization landscape with fundamental degradation physics, thereby ensuring predictions are physically plausible even in data-sparse regimes.

The report presents a comprehensive review of the state-of-the-art, identifying a critical gap in lightweight hybrid modeling suitable for edge-compatible architecture in Battery Management Systems (BMS). Mathematically formulates the SOH estimation problem, defines the governing physical constraints, and outlines a rigorous methodology for validation against benchmark datasets from the NASA Prognostics Center of Excellence (PCoE) and the Center for Advanced Life Cycle Engineering (CALCE). Theoretical analysis and architectural design suggests that this hybrid approach will significantly improve data efficiency and cross-chemistry generalization compared to purely data-driven baselines, offering a viable pathway for robust onboard diagnostics.

CONTENTS

	Page
Acknowledgements	i
ABSTRACT	ii
LIST OF TABLES	vi
LIST OF FIGURES	viii
ABBREVIATIONS	xi
NOTATION	xiii
CHAPTER 1 INTRODUCTION	1
1.1 The Global Context of Battery Prognostics	1
1.2 The Degradation Mechanism and Modeling Challenges	2
1.3 The Rise of Data-Driven and Hybrid Approaches	4
1.4 Problem Statement and Research Gap	6
1.5 Research Objectives	6
1.6 Organization of the Report	7
CHAPTER 2 LITERATURE REVIEW	9
2.1 Model-Based Approaches: The Foundation of Interpretability	9
2.1.1 Electrochemical Models (White-Box)	9
2.1.2 Empirical Models (Grey-Box)	10
2.2 Data-Driven Approaches: The Accuracy Frontier	11
2.2.1 Recurrent Neural Networks (RNNs) and Variants	11
2.2.2 The Transformer Revolution	12
2.3 Hybrid Physics-Informed Frameworks: The Convergence	13
2.3.1 PDE-Based PINNs (The “Heavy” Approach)	13
2.3.2 ODE-Regularized Networks (The “Lightweight” Approach)	14
2.4 Gap Analysis and Opportunity Statement	15
CHAPTER 3 PROBLEM DEFINITION AND FORMULATION	16
3.1 Problem Definition	16
3.2 Physics Formulation (The Regularizer)	17
3.2.1 The Verhulst (Logistic) Model	17
3.2.2 The Double-Exponential Model	18
3.3 The Optimization Objective	19

CHAPTER 4	METHODOLOGY	21
4.1	System Architecture	21
4.2	Deep Learning Component: Gated Recurrent Unit (GRU)	22
4.3	Adaptive Loss Balancing (Uncertainty Weighting)	24
4.4	Two Physics Routes and What Is Compared	25
CHAPTER 5	DATASETS, EXPERIMENTAL SETUP, AND BASELINES	26
5.1	Dataset Suitability and Defense	26
5.1.1	Technical Data Breakdown and Preprocessing	29
5.2	Evaluation Strategy	35
5.3	Benchmarking Baselines	36
5.4	Resource Feasibility	38
CHAPTER 6	RESULTS AND DISCUSSION	41
6.1	Evaluation Protocol	41
6.2	Physics Fits and Interpretability	42
6.3	Main Benchmark (NASA Leave-One-Cell-Out)	44
6.4	The Central Finding: Learnable Physics (Route B) Outperforms the Fixed Prior	47
6.5	The Transformer Question: Heavy Attention Does Not Pay on Scarce Data	50
6.5.1	The Paired Gap	50
6.5.2	Why the Heavy Transformer Fails Here: Overfitting Under Scarcity . .	51
6.5.3	Cross-Dataset Nuance	53
6.6	Prognostic (PHM) Assessment	54
6.7	Ablation Studies	56
6.8	Deployment Footprint (Cortex-M4)	58
6.9	Implementation Caveats	60
6.10	Summary of Findings	61
CHAPTER 7	CONCLUSION AND FUTURE WORK	63
7.1	Conclusion	63
7.2	Future Work	65
APPENDIX A	CLASSIFICATION OF REVIEWED LITERATURE	66
BIBLIOGRAPHY		67

LIST OF TABLES

Table	Caption	Page
5.1	Specification and Characteristics of Selected Datasets	27
5.2	Detailed Feature Breakdown of Processed Datasets	30
5.3	Benchmark Baselines Comparison Matrix	37
5.4	Deployment configuration for the lightweight hybrid (parameters, RAM and MACs measured; int8 size and latency are the clearly-labelled estimates described above). Full per-model figures appear in Table 6.11.	39
6.1	Closed-form degradation priors fit per cell (real, measured). The logistic carrying capacity K is cell-specific and exceeds 1.0 for the NASA cells, consistent with the $>100\%$ initial SOH allowed by the $\text{SOH} = Q/Q_{\text{rated}}$ convention.	42
6.2	Model benchmark on NASA leave-one-cell-out (real measured values; SOH RMSE as mean $\pm$ 95% CI). “Phys.Cons.” is the fraction of predictions respecting $d\text{SOH}/dt \leq 0$; “Cross” is the zero-shot NASA→CALCE RMSE. The first two rows are the non-learned reference floors of Chapter 5 (Table 5.3): the empirical double-exponential curve fit (fit on the training cells, evaluated on the held-out cell) and a non-recurrent SVR on the same windows. The two hybrids use a deliberately generous hidden size for this comparison; the matched-size (11k-parameter) deployment model reproduces the route-B accuracy (Section 6.8).	45
6.3	Per-cell SOH RMSE (mean over 3 seeds) under leave-one-cell-out. The fixed prior (route A) and the purely data-driven LSTM are excellent on the typical cells (B0007, B0018) but degrade sharply on the atypical B0005/B0006; the learnable route B is uniformly robust across all four.	48
6.4	Hybrid vs. the heavy (SOTA-scale) Transformer, paired at the <i>fold</i> level ($n = 12$), with the paired Wilcoxon signed-rank test.	50
6.5	Data-scarcity stress test: test SOH RMSE as each training cell is subsampled to 10–100% of its cycles (route-A hybrid used as the inexpensive hybrid representative). The physics-regularised hybrid is essentially flat across a $10\times$ change in data volume; the heavy Transformer is at its <i>worst</i> at full data, and the light Transformer is erratic throughout.	52
6.6	Generalisation gap (test – train RMSE) at full data. Scaling the Transformer $13\times$ increases its gap by half again and yields the worst test error of the three; the hybrid’s nominally larger gap is measured from a far lower train error and still ends at the best test RMSE.	52
6.7	Prognostic (PHM) metrics on NASA leave-one-cell-out (mean over folds $\times$ seeds). CRA punishes early-life over- and under-shoot; negative CRA indicates badly calibrated early-life horizons.	54

6.8	Physics-design ablation (route, prior, weighting), with a de-confound control. “Physics OFF + t-channel” has the route-B architecture (and thus the extra cycle-coordinate input) but no ODE-residual penalty, isolating the input feature from the physics.	56
6.9	Aggregation ablation (route-A hybrid with double-exponential prior, mean over 2 folds $\times$ 2 seeds).	57
6.10	Data-volume ablation (route-A hybrid, stratified per-cell subsample, mean over 2 folds $\times$ 2 seeds).	58
6.11	Deployment footprint. Parameter counts, fp32 sizes, MACs and activation RAM are measured on the trained models (state-dict serialisation and forward hooks); the int8 column is the standard 4 $\times$ post-training-quantisation estimate of the fp32 weight size, and the Cortex-M4 latency is an analytic MACs/clock (CMSIS-NN-style) estimate. These supersede the placeholder estimates of the earlier report.	59
A.1	Methodological Classification of Cited Works	66

LIST OF FIGURES

Figure	Caption	Page
1.1	Typical Li-ion battery degradation trajectory illustrating the two dominant aging mechanisms: Linear decay driven by SEI growth (LLI) and accelerated aging (knee-point) driven by Loss of Active Material (LAM).	4
4.1	Schematic of the proposed Hybrid Physics-Informed Framework. The architecture integrates Savitzky-Golay smoothing for feature stability and utilizes a parallel physics regularizer (ODE) to constrain the GRU network via an uncertainty-weighted composite loss function.	22
4.2	Internal structure of the Gated Recurrent Unit (GRU). Unlike LSTMs which utilize three gates, the GRU employs only two (Update and Reset), reducing parameter count and memory footprint, making it highly suitable for edge-compatible BMS implementations.	23
5.1	NASA PCoE capacity fade for all four LCO 18650 cells over their true 132–168 cycle lifetimes. Regeneration “jumps” after rest periods are visible as upward spikes; the dashed line marks the 80% EOL (1.60 Ah) against the 2.0 Ah rated capacity.	28
5.2	CALCE CS2_35 candidate timelines (same physical cell, aligned by date). The <code>channel_full</code> timeline (878 cycles from 2010-08-16, 1.14→0.30 Ah) is adopted; the <code>statistics</code> export is the same cell minus early life and is not used. The knee-point and 80% EOL are marked.	29
5.3	Per-cell correlation between each engineered feature and SOH. The ICA magnitude, ICA area, discharge time and mean voltage are near-perfectly correlated with health across all NASA cells and CALCE CS2_35, while intra-cycle voltage variance and the IR proxy carry complementary (negative) information.	31
5.4	NASA B0005 capacity with detected regeneration jumps (markers). These upward “healing” artifacts violate monotonic degradation and motivate the physics-consistency constraint enforced by the hybrid model.	32
5.5	CALCE CS2_35 (<code>channel_full</code>) raw vs. Savitzky–Golay smoothed capacity over its full 878-cycle life (1.14 → 0.30 Ah). The filter (window 11, polyorder 3) removes the sharp downward spikes from reference/impedance cycles while preserving the underlying degradation trend.	33
5.6	CALCE capacity extraction. (Left) Taking the <i>maximum</i> of the cumulative <code>Discharge_Capacity</code> produces spurious values up to ~54 Ah — the source of the earlier report’s “48–54 Ah” outliers. (Right) The correct per-cycle increment (ΔQ) yields a clean degradation curve with only 4 true outliers outside the plausible [0.30, 1.30] Ah band.	34

6.1	Closed-form prior comparison per cell. (Left) the double-exponential attains a higher fit quality (R^2) than the logistic on every cell; (Right) it also produces a smaller residual spread. This justifies retaining the two-timescale prior alongside the logistic.	43
6.2	Closed-form prior fits for the atypical cell B0006. The cell starts above rated capacity and regenerates heavily; the fixed closed-form prior (route A) cannot track this without dragging predictions away from the data.	44
6.3	SOH RMSE on NASA leave-one-cell-out (mean $\pm$ 95% CI; parameter counts above the bars). The two leftmost bars are the non-learned floors (double-exponential curve fit, SVR); every learned model beats them. The route-B hybrid attains the lowest mean RMSE of all models, the LSTM is the strongest purely data-driven baseline, and <i>both</i> Transformers — including the 225k-parameter SOTA-scale one — trail the far smaller recurrent models.	46
6.4	Physical plausibility (fraction of predictions with $d\text{SOH}/dt \leq 0$). The route-B hybrid is the most physically consistent <i>learned</i> model (0.939), well above all purely data-driven baselines. The double-exponential curve fit scores higher (0.995) only trivially — a fitted closed-form curve is monotone by construction, which is also why it cannot track genuine regeneration.	46
6.5	SOH trajectory on the held-out cell B0005. Every model — including the fixed-prior route A — sits <i>above</i> the measured curve, over-estimating SOH as the real capacity fades faster than a prior fit on the other cells predicts. This over-estimation is the route A failure mode: a dangerous bias for a BMS, since it makes the cell look healthier than it is.	49
6.6	Differential error plot on B0018: per-cycle $\Delta \text{SOH error} $ (baseline – hybrid). Values above zero indicate the hybrid is closer to ground truth than the baseline at that cycle.	51
6.7	Data scarcity drives accuracy more than model size. (Left) test RMSE vs. training-data fraction: the physics-regularised hybrid is flat, the heavy Transformer degrades at full data. (Middle) train vs. test curves — the gap is the overfitting. (Right) in the extreme low-data regime (10%), the hybrid beats both Transformers despite the heavy one having $5\times$ its parameters.	53
6.8	α - λ RUL prognosis for the hybrid on the held-out cell B0018. The predicted RUL stays within (or close to) the $\pm 20\%$ α -cone through most of the cell’s degradation, converging to the true end-of-life.	55
6.9	SOH trajectory on the held-out cell B0018. The hybrid tracks the measured degradation, including capacity-regeneration peaks, without the phase lag visible in the purely data-driven baselines.	55
6.10	Cortex-M4 footprint comparison: int8 flash (log scale, against the 1 MB budget), working RAM, and analytic latency at 80 MHz. The deployable route-B hybrid needs 1/20 the flash and 1/11 the latency of the SOTA-scale Transformer, whose fp32 form (931 KB) barely fits the flash budget at all.	59

6.11	The accuracy-vs-memory frontier. (Left) the deployed 11k-parameter hybrid occupies the favourable lower-left corner: more accurate than both Transformers at a fraction of the size. (Right) the deployed hybrid relative to the SOTA-scale Transformer on every axis: $0.71\times$ the error at $0.05\times$ the parameters and flash and $0.09\times$ the latency.	60
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ABBREVIATIONS

AI	Artificial Intelligence.
ANN	Artificial Neural Network.
BMS	Battery Management System.
CALCE	Center for Advanced Life Cycle Engineering.
DL	Deep Learning.
DVA	Differential Voltage Analysis.
EOL	End of Life.
EV	Electric Vehicle.
GRU	Gated Recurrent Unit.
ICA	Incremental Capacity Analysis.
ICE	Internal Combustion Engine.
IITM	Indian Institute of Technology Madras.
LAM	Loss of Active Material.
Li-ion	Lithium-ion.
LLI	Loss of Lithium Inventory.
LSTM	Long Short-Term Memory.
MAE	Mean Absolute Error.
MSE	Mean Squared Error.
NASA PCoE	NASA Prognostics Center of Excellence.
ODE	Ordinary Differential Equation.
PDE	Partial Differential Equation.

PIML	Physics-Informed Machine Learning.
PINN	Physics-Informed Neural Network.
RMSE	Root Mean Squared Error.
RNN	Recurrent Neural Network.
RUL	Remaining Useful Life.
SEI	Solid Electrolyte Interphase.
SOH	State of Health.

NOTATION

$\hat{y}$	Predicted State of Health
λ_{phys}	Weighting factor for the physics loss term
$\mathcal{L}_{total}$	Composite Loss Function
σ_{data}	Aleatoric uncertainty (observation noise) for data
σ_{phys}	Aleatoric uncertainty for physics equation
θ	Trainable parameters of the Neural Network
θ_{phys}	Learnable parameters of the physics ODE (a, b, c, d)
a, b	Parameters for the fast decay process (Double-Exponential Model)
c, d	Parameters for the slow decay process (Double-Exponential Model)
dQ/dV	Incremental Capacity Analysis (ICA) curve
h_t	Hidden state vector of the GRU
I	Battery Current
K	Carrying capacity/Saturation limit (Verhulst Model)
$Q_{max,k}$	Maximum releasable capacity at cycle k
Q_{rated}	Nominal rated capacity of the battery
r	Intrinsic degradation rate constant (Verhulst Model)
r_t	Reset gate of the GRU
RUL	Remaining Useful Life
SOH_k	State of Health at cycle k

T	Temperature
V	Battery Terminal Voltage
x_t	Input feature vector at time step t
z_t	Update gate of the GRU

CHAPTER 1

INTRODUCTION

1.1 THE GLOBAL CONTEXT OF BATTERY PROGNOSTICS

The transition from internal combustion engines (ICE) to electric propulsion is one of the major industrial shifts of the 21st century. As nations work toward the climate goals of the Paris Agreement and subsequent COP summits, the demand for high-performance, reliable energy storage systems has grown rapidly.

Lithium-ion (Li-ion) batteries have emerged as the dominant technology for this transition due to their high energy density, low self-discharge rates, and lack of memory effect compared to predecessors like Nickel-Metal Hydride (NiMH) or Lead-Acid batteries Madani *et al.* (2025).

However, these electrochemical systems are complex, time-variant, and non-linear. They degrade inevitably with usage and time, a process influenced by a multitude of stress factors including ambient temperature, charge/discharge rates (C-rates), Depth of Discharge (DOD), and mechanical vibrations NASA Prognostics Center of Excellence (PCoE) (2025). This degradation is not merely a performance issue regarding vehicle range but a critical safety and economic concern.

The State of Health (SOH), typically defined as the ratio of the battery's current maximum capacity to its rated capacity, is the primary metric for determining the viability of a battery pack. When SOH drops below a certain threshold—typically 70–80% for automotive applications—the battery is considered “retired” from vehicular service due to the increased risk of failure and reduced range capability Ma *et al.* (2025). The Remaining Useful Life (RUL) predicts the number of cycles or time remaining until this threshold is reached.

Inaccurate estimation of these parameters leads to two distinct failure modes in battery management, each with severe consequences:

- **Over-estimation of RUL:** This poses severe safety risks. If a BMS believes a battery is healthier than it is, it may allow aggressive charging or discharging protocols that the degraded cell cannot handle, leading to lithium plating, internal short circuits, or in extreme cases, thermal runaway and fire Madani *et al.* (2025). Additionally, it contributes to unexpected vehicle stranding (“range anxiety”), which erodes consumer confidence in EV technology.
- **Under-estimation of RUL:** This results in the premature disposal or recycling of viable assets. If a battery is retired at 85% health because the BMS incorrectly estimates it at 75%, the total cost of ownership for the vehicle increases significantly. Furthermore, this premature retirement contributes to unnecessary environmental waste and supply chain strain for critical minerals like lithium, cobalt, and nickel Madani *et al.* (2025).

Use-Case Clarification: To make the engineering objective explicit: the proposed framework targets onboard Battery Management System (BMS) usage for passenger EVs and light commercial vehicles, as well as for fleet managers requiring robust SOH/RUL estimates for operational decisions (charging strategy, warranty, second-life planning). The model is designed to be trained offline (server/GPU) and deployed as a compact inference model on edge hardware. This separation—offline training, online lightweight inference—is core to the design decisions made in this thesis, specifically the choice of a GRU backbone and ODE regularizers. The practical benefits are safer charging protocols, improved range estimation, and reduced premature retirement of battery packs.

The development of accurate and computationally feasible prognostic algorithms is therefore a practical requirement for the large-scale adoption of electric mobility, not only an academic problem.

1.2 THE DEGRADATION MECHANISM AND MODELING CHALLENGES

To understand the difficulty of the prognostic challenge, one must appreciate the underlying electrochemistry that the algorithms attempt to model. Battery aging is governed by two primary, coupled mechanisms: the *Loss of Lithium Inventory (LLI)* and the *Loss of*

Active Material (LAM) Madani et al. (2025).

Loss of Lithium Inventory (LLI) is primarily caused by the formation and continuous thickening of the Solid Electrolyte Interphase (SEI) layer on the anode during charging. The SEI layer is a passivation film that forms from the decomposition of the electrolyte. While essential for stability, its continuous growth consumes cyclable lithium ions, permanently removing them from the energy storage process. This mechanism is often dominant in the early stages of battery life and typically follows a linear or square-root-of-time trajectory Xue *et al.* (2023).

Loss of Active Material (LAM) involves the structural degradation of the electrode materials. This includes phenomena such as particle cracking, binder decomposition, and electrical isolation of active material sites. LAM is often exacerbated by the mechanical stress of repeated intercalation and de-intercalation of ions, which causes volume expansion and contraction of the electrode lattice. LAM is frequently responsible for the accelerated aging phase or “knee-point” observed in the later stages of a battery’s life, where capacity drops precipitously after a period of linear decay Madani *et al.* (2025).

Capturing this non-linearity—specifically the transition from linear degradation to the accelerated knee-point—is the central challenge of modern Battery Management Systems (BMS). Traditional industrial BMS rely on Equivalent Circuit Models (ECMs) or simple Coulomb counting. These methods treat the battery as a linear capacitor-resistor network. While computationally inexpensive, they fail to capture the complex physicochemical changes occurring during aging, leading to significant accumulation of error over time Naseri *et al.* (2023). They are essentially “blind” to the internal state of health until a calibration event occurs.

Conversely, high-fidelity electrochemical models, such as the Doyle-Fuller-Newman (DFN) model or the Pseudo-Two-Dimensional (P2D) model, describe the internal ion

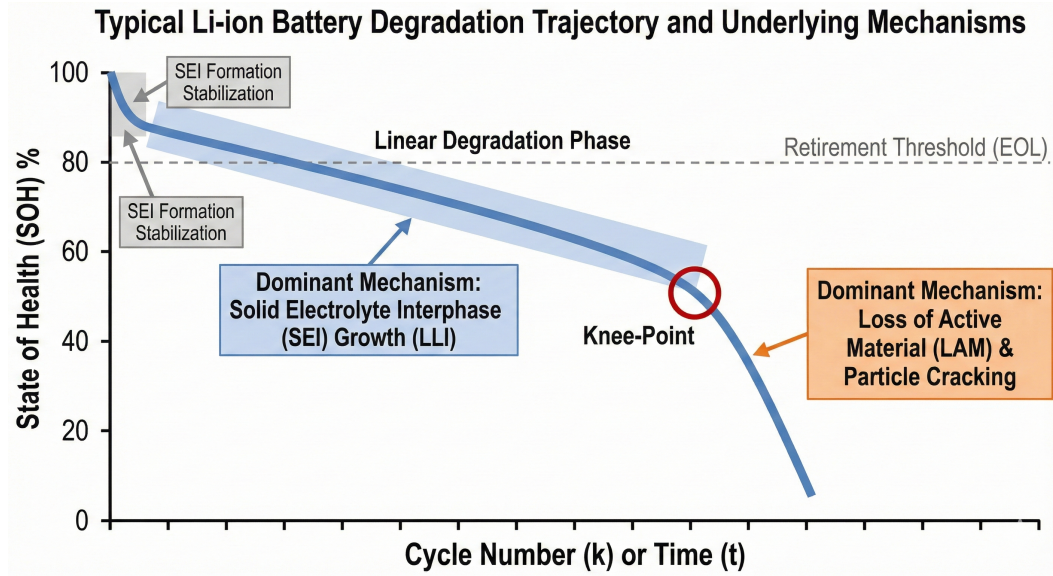


Figure 1.1: Typical Li-ion battery degradation trajectory illustrating the two dominant aging mechanisms: Linear decay driven by SEI growth (LLI) and accelerated aging (knee-point) driven by Loss of Active Material (LAM).

concentrations, potentials, and intercalation kinetics using coupled Partial Differential Equations (PDEs) Wen *et al.* (2024). While highly accurate and scientifically interpretable, these “white-box” models are computationally prohibitive for the low-power microcontrollers found in standard EVs. Furthermore, they require the identification of over 20 specific electrochemical parameters (e.g., diffusion coefficients, reaction rate constants, pore tortuosity), many of which can only be determined through destructive testing. This renders them impractical for generalized use across different battery chemistries or for aging batteries where these parameters drift over time Kajiura (2025).

1.3 THE RISE OF DATA-DRIVEN AND HYBRID APPROACHES

The limitations of purely physical models have catalyzed a massive shift toward “black-box” data-driven approaches in the academic literature. With the advent of large-scale open-source datasets from institutions like the NASA Prognostics Center of Excellence (PCoE) and the Center for Advanced Life Cycle Engineering (CALCE), researchers have

demonstrated that deep learning architectures can map voltage, current, and temperature inputs directly to SOH outputs with remarkable accuracy Onyenagubo (2025). Models like Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) have become the standard for handling time-series battery data due to their ability to capture temporal dependencies and history-dependent degradation Zhang *et al.* (2025).

However, purely data-driven models suffer from three fundamental flaws that limit their industrial adoption:

1. **Lack of Interpretability:** They operate as “black boxes,” offering no insight into *why* a battery is degrading. A fleet operator cannot determine if a capacity drop is due to recoverable conditions (like low temperature) or irreversible aging.
2. **Physical Inconsistency:** A deep learning model trained on noisy data might learn to predict that a battery’s capacity spontaneously increases (self-healing), violating the second law of thermodynamics. Such predictions reduce user trust and can confuse BMS logic Madani *et al.* (2025).
3. **Data Hunger and Generalization:** These models require massive labeled datasets to generalize effectively. In real-world scenarios, “run-to-failure” data is scarce because vehicles are rarely driven until the battery completely dies. Furthermore, models trained on one battery chemistry or load profile often fail catastrophically when applied to another, known as the “domain shift” problem Tran (2025).

This thesis develops a **Hybrid** approach that occupies the optimal middle ground. By integrating physical laws into the deep learning optimization process—a technique known as *Physics-Informed Machine Learning (PIML)*—the framework is both accurate and physically consistent. Specifically, it is a “Physics-Regularized” framework in which the neural network is constrained by simple, empirically derived Ordinary Differential Equations (ODEs) rather than complex PDEs. This choice is strategic, aiming to retain the interpretability benefits of PIML while ensuring the computational feasibility required for real-world edge deployment Jafari and Byun (2023).

1.4 PROBLEM STATEMENT AND RESEARCH GAP

While the literature abounds with battery prognostic methods, a critical analysis reveals specific gaps that this thesis addresses.

Most existing **Physics-Informed Neural Networks (PINNs)** in the battery domain focus on solving cell-level electrochemistry (e.g., the Single Particle Model) using neural networks as surrogate PDE solvers Xue *et al.* (2023). While scientifically rigorous, these approaches are often *more* computationally expensive to train than standard data-driven models due to the calculation of spatial derivatives and residuals at every step Wong (2025). There is a distinct lack of research on “lightweight” PINNs that utilize simple, empirical ODEs (like the Verhulst or Double-Exponential models) to regularize standard RNNs for system-level prognostics.

Furthermore, existing hybrid models often use static weights to balance the data loss and physics loss, which requires exhaustive manual tuning. The application of dynamic loss balancing strategies, such as uncertainty weighting, remains under-explored in this specific context.

Therefore, the core problem this thesis addresses is: **How to estimate SOH and RUL with high accuracy and physical consistency on resource-constrained hardware, given limited and noisy training data?**

1.5 RESEARCH OBJECTIVES

The primary objective of this thesis is to develop and validate a hybrid deep learning framework for battery prognostics that outperforms traditional methods in terms of data efficiency, physical plausibility, and computational feasibility. The specific objectives of this thesis are:

- **Comprehensive Literature Review:** To systematically analyze the state-of-the-art in SOH/RUL estimation, identifying the specific limitations of current PINN implementations (which rely heavily on computationally expensive PDEs) and

establishing the theoretical basis for ODE-based regularization.

- **Hybrid Architecture Development:** To design a novel deep learning architecture that integrates a Gated Recurrent Unit (GRU) network with empirical degradation models (Verhulst and Double-Exponential) via a composite loss function. This objective specifically addresses the need for a “lightweight” PINN suitable for edge computing.
- **Advanced Loss Formulation:** To implement a dynamic loss weighting strategy based on homoscedastic uncertainty (Bayesian approximation) that automates the balancing of “data loss” and “physics loss,” and to test it empirically against static weighting (Chapter 6).
- **Rigorous Benchmarking:** To establish performance baselines using purely model-based methods (curve fitting) and purely data-driven methods (Standard GRU, SVR). This comparative analysis is crucial for quantifying the specific value-add of the hybrid approach, addressing the feedback regarding the necessity of model-driven baselines.
- **Validation on Diverse Datasets:** To validate the model’s robustness and generalizability using distinct datasets (NASA PCoE for constant loading, CALCE for dynamic loading), focusing on the model’s ability to handle data artifacts like “capacity regeneration” without making physically impossible predictions.
- **Quantification of Computational Cost:** To evaluate the inference time and parameter count of the proposed hybrid model against Transformer baselines, ensuring suitability for resource-constrained Battery Management Systems (BMS)

1.6 ORGANIZATION OF THE REPORT

The remainder of this thesis is organized as follows.

Chapter 2 presents an exhaustive review of the literature , categorizing existing methodologies into model-based, data-driven, and hybrid paradigms, and identifies specific research gaps.

Chapter 3 provides the formal problem definition, detailing the mathematical formulation of SOH and the physics-informed constraints.

Chapter 4 outlines the theoretical methodology, including the GRU architecture and the

uncertainty-weighted loss function.

Chapter 5 analyzes the datasets and the feature engineering strategy required to extract meaningful health indicators, and defines the evaluation protocol and baselines.

Chapter 6 presents the measured experimental results: the model benchmark, the central finding on learnable vs. fixed physics, the quantified gap to the SOTA-scale Transformer (including the data-scarcity and overfitting stress tests), the prognostic (PHM) assessment, the ablation studies, and the deployment footprint.

Chapter 7 concludes the thesis and outlines future work.

CHAPTER 2

LITERATURE REVIEW

The field of battery prognostics is rapidly evolving, characterized by a transition from empirical correlations to sophisticated AI-driven frameworks. This chapter reviews key contributions to the field, categorizing them into three dominant paradigms: Model-Based, Data-Driven, and Hybrid approaches. The review critically analyzes the strengths and limitations of each paradigm to contextualize the proposed hybrid framework and defend the methodological choices made for this thesis.

2.1 MODEL-BASED APPROACHES: THE FOUNDATION OF INTERPRETABILITY

Model-based approaches are grounded in the explicit mathematical description of battery behavior. They are historically significant as they provide deep insights into the internal states of the cell.

2.1.1 Electrochemical Models (White-Box)

The gold standard for physical fidelity is the electrochemical model, often described by the **Doyle-Fuller-Newman (DFN)** framework or the **Single Particle Model (SPM)**. These models utilize coupled Partial Differential Equations (PDEs) to describe lithium-ion concentration gradients, potential distributions, and electrolyte dynamics. Naseri et al. (2023) and Xue et al. (2023) demonstrated that these models can achieve high prognostic accuracy by simulating the growth of the SEI layer. They provide a transparent view of internal degradation, allowing engineers to pinpoint whether capacity fade is due to anode degradation or electrolyte depletion.

Limitations: However, the computational cost of solving these PDEs at every time

step is prohibitive for real-time applications. Solving the P2D model typically requires solving a system of differential-algebraic equations with thousands of degrees of freedom. Furthermore, they require the identification of over 20 specific electrochemical parameters (e.g., diffusion coefficients, reaction rate constants, pore tortuosity), many of which can only be determined through destructive testing Kajiura (2025). This renders them impractical for general-purpose BMS that must manage cells with manufacturing variances. As highlighted in feedback, pure PINN approaches relying on these PDEs are often too computationally expensive for real-world embedded use Madani *et al.* (2025).

2.1.2 Empirical Models (Grey-Box)

To circumvent the complexity of PDEs, researchers have developed empirical models that approximate degradation using simplified differential equations.

The Verhulst (Logistic) Model: Originally developed for population dynamics, the Verhulst model was adapted for battery prognostics by Xian et al. Xian *et al.* (2014). It posits that battery degradation follows an S-curve, limited by a “carrying capacity” (the total active material).

$$\frac{dSOH}{dt} = r \cdot SOH \left(1 - \frac{SOH}{K} \right) \quad (2.1)$$

Analysis: This model effectively captures the saturation effect of degradation. Its mathematical simplicity makes it an ideal candidate for the “physics regularization” term in this project. However, its rigid S-shape often fails to capture the multi-stage degradation observed in modern chemistries, particularly the transition from linear aging to the accelerated “knee-point” Madani *et al.* (2025). Despite this, when coupled with a neural network, the Verhulst model can serve as a robust boundary constraint.

The Double-Exponential Model: To address the rigidity of the logistic model, Ma et al. Ma *et al.* (2025) proposed a Double-Exponential model ($y = a \cdot e^{bx} + c \cdot e^{dx}$).

Mechanism: This formulation assumes degradation is the sum of two distinct decay processes: a fast initial decay (SEI formation stabilization) and a slow long-term decay

(active material loss). **Significance:** This model has become a robust baseline for curve-fitting approaches due to its flexibility in fitting the “knee-point.” Establishing this as a baseline in this thesis is crucial to demonstrate whether the hybrid AI model offers superior accuracy over this sophisticated empirical fit Xue *et al.* (2023). It directly answers the feedback regarding the need for a model-driven baseline.

Particle Filters (PF): While not a model per se, Particle Filters are often used in conjunction with empirical equations to estimate state variables under uncertainty Ahwiadi and Wang (2024). While effective for tracking non-linear systems, PFs suffer from “particle degeneracy” and require significant computational resources to maintain the probability distribution of the state (often needing hundreds of particles), making them heavier than simple ODE integration.

Note on Classification: Each referenced work in the bibliography has been categorized by approach (PDE-PINN, empirical ODE, particle filter, or data-driven) in the Appendix. This explicit mapping clarifies that most recent PINN work in batteries attempts to embed high-fidelity PDEs; ODE-regularized hybrids remain a small subset, and within it no prior work compares fixed against *learnable* ODE parameters under a leave-one-cell-out protocol, or validates uncertainty weighting against a tuned fixed weight — the two comparisons this thesis contributes.

2.2 DATA-DRIVEN APPROACHES: THE ACCURACY FRONTIER

The availability of high-quality open-source datasets has fueled the dominance of data-driven methods, which treat the battery as a “black box” and learn the mapping from inputs to SOH.

2.2.1 Recurrent Neural Networks (RNNs) and Variants

Standard Feed-Forward Networks (FFNs) are unsuitable for prognostics as they cannot capture temporal dependencies. The field rapidly shifted to Recurrent Neural Networks

(RNNs) and their advanced variants.

LSTM vs. GRU: Zhang et al. Zhang *et al.* (2025) and others established LSTMs as a strong baseline. However, recent comparative studies by Li et al. Jafari and Byun (2023) indicate that Gated Recurrent Units (GRUs) often match or exceed LSTM performance with lower computational cost (two gates vs. three).

Relevance to Thesis: This efficiency makes the GRU an ideal candidate for the “lightweight” framework proposed in this thesis. The GRU’s ability to model long-term dependencies with fewer parameters aligns with the goal of edge deployment.

CNN-RNN Hybrids: Jafari et al. Jafari and Byun (2023) demonstrated that 1D-Convolutional Neural Networks (CNNs) are effective at extracting local features from charging curves, which are then passed to GRUs for temporal forecasting. This hierarchical feature-extraction principle informs the physics-informed feature engineering (ICA/DVA) adopted in this work.

2.2.2 The Transformer Revolution

The current state-of-the-art in predictive accuracy involves Transformer models, which utilize self-attention mechanisms to capture global dependencies in time-series data.

SGEformer: Fauzi et al. Fauzi *et al.* (2024) introduced the SGEformer (Seasonal and Growth Embedding Transformer), which achieves exceptionally low error rates (MSE 0.0001) on NASA datasets.

Limitations: Despite their accuracy, Transformers have a quadratic memory complexity ($O(L^2)$) with respect to sequence length, making them difficult to deploy on resource-constrained hardware compared to the linear complexity ($O(L)$) of RNNs Onyenagubo (2025). Equally important, their reported accuracy is obtained on large training corpora: self-attention models require substantial data, and in low-data regimes their excess capacity leads to overfitting rather than higher accuracy. This matters for the BMS setting,

where run-to-failure data for a given cell design is scarce; this thesis tests the point empirically by benchmarking two Transformer scales at realistic data volumes (Chapter 6). Furthermore, purely data-driven models like SGEformer are prone to overfitting to dataset artifacts, such as sensor noise masquerading as capacity regeneration, leading to physically implausible predictions (e.g., predicting infinite battery life in certain local minima).

Comparative Note: For practical on-device feasibility, GRU-based hybrid solutions represent a favourable accuracy–cost frontier; Transformers yield strong accuracy but with an unfavorable compute/memory trade-off for on-device deployment.

2.3 HYBRID PHYSICS-INFORMED FRAMEWORKS: THE CONVERGENCE

Physics-Informed Machine Learning (PIML) represents the frontier of battery prognostics, aiming to merge the interpretability of physics with the flexibility of deep learning.

2.3.1 PDE-Based PINNs (The “Heavy” Approach)

The seminal work in PINNs involves using neural networks to solve the governing PDEs of a system. In the battery domain, Xue et al. Xue *et al.* (2023) applied PINNs to the Single Particle Model (SPM).

Critique: While this approach retains rigorous physical fidelity, it does not solve the computational bottleneck. Computing the spatial derivatives (gradients) required for the PDE residuals via automatic differentiation is computationally expensive during the training phase and often during inference if the physics loss is evaluated for online adaptation Wong (2025). This aligns with the “Resource Requirement” feedback, which questioned the feasibility of pure PINNs for real-world use. The computational cost of these PDE-PINNs makes them infeasible for the specific M.Tech project constraints identified in the feedback.

2.3.2 ODE-Regularized Networks (The “Lightweight” Approach)

A more pragmatic branch of PIML utilizes simpler Ordinary Differential Equations (ODEs) as regularizers. Wen et al. Wen *et al.* (2024) pioneered a framework fusing a neural network with the Verhulst ODE for battery prognostics. They employed an uncertainty-based weighting method to balance the data loss and physics loss, demonstrating improved generalization. Earlier, Xian et al. Xian *et al.* (2014) had established the Verhulst law itself as a viable degradation model within a particle-filter framework.

Remaining Gaps: However, significant gaps remain. Most existing hybrid models rely on raw voltage and current inputs rather than advanced physics-informed features (like Incremental Capacity Analysis - ICA). Additionally, they have largely restricted themselves to the Verhulst model, ignoring more flexible priors like the Double-Exponential model which are better suited for “knee-point” prediction.

Novelty Statement: While prior studies have embedded ODE priors into neural networks, this research addresses a distinct gap in the literature by integrating four specific components into a unified framework: (a) a Gated Recurrent Unit (GRU) backbone selected for edge-compatibility; (b) the use of Double-Exponential priors alongside Verhulst models to capture accelerated “knee-point” degradation, in both a *fixed* closed-form variant and a *learnable* ODE-residual variant whose parameters train jointly with the network; (c) a systematic empirical evaluation of homoscedastic uncertainty weighting against fixed loss weights in this setting — a comparison largely absent from the battery literature; and (d) physics-informed ICA/DVA feature engineering. This specific architectural combination, together with its measured edge-deployment profile, constitutes the primary novelty of this thesis, offering a practical solution that bridges the gap between computationally expensive physical models and purely data-driven approaches.

Comparison with Closest Prior Works. Among the reviewed literature, the closest approaches are Wen et al. Wen *et al.* (2024), who employed a Verhulst-law-based ODE regularization with uncertainty weighting, and Xue et al. Xue *et al.* (2023), who applied PINNs with the full electrochemical SPM model. Wen et al. did not compare fixed against learnable physics parameters nor include Double-Exponential degradation priors, and their method was not designed for edge deployment. Xue et al. demonstrated high-fidelity PDE-PINNs, but their computational cost and memory footprint make them unsuitable for on-device BMS inference. In contrast, the present work combines lightweight GRU encoders with simplified ODE-based physics priors, integrates ICA/DVA features, and empirically evaluates both fixed and uncertainty-based loss weighting while explicitly constraining model complexity for MCU deployment. This positions the proposed approach as a simplified and computationally practical alternative to existing PINN and hybrid methods.

2.4 GAP ANALYSIS AND OPPORTUNITY STATEMENT

The literature review reveals distinct gaps that this thesis aims to fill:

- **Computational Feasibility Gap:** Current literature is dominated by PDE-based PINNs which are rigorous but infeasible on standard BMS hardware. There is a distinct lack of research on “lightweight” PINNs that utilize simple empirical ODEs for system-level prognostics.
- **Loss Balancing Gap:** Static weights for physics vs. data loss are brittle and require manual tuning. Dynamic strategies like homoscedastic uncertainty weighting remain under-explored in battery prognostics.
- **Feature Gap:** Many hybrid works rely on raw voltage/current signals; integrating physics-informed features (like ICA/DVA) into a hybrid pipeline offers untapped robustness.
- **Benchmark Gap:** Comparisons often omit a rigorous model-driven baseline (Double-Exponential Curve Fitting). This thesis includes this baseline to rigorously isolate the value-add of the AI component.

CHAPTER 3

PROBLEM DEFINITION AND FORMULATION

This chapter provides a rigorous mathematical definition of the SOH estimation problem and formulates the specific physical models that govern the hybrid framework.

3.1 PROBLEM DEFINITION

The objective of the thesis is to estimate the State of Health (*SOH*) of a Lithium-ion battery at a given cycle index k , based on a sequence of historical measurements. State of Health (*SOH*) is defined as the ratio of the maximum releasable capacity at the current cycle ($Q_{max,k}$) to the nominal rated capacity (Q_{rated}) provided by the manufacturer:

$$SOH_k = \frac{Q_{max,k}}{Q_{rated}} \times 100\% \quad (3.1)$$

The problem is modeled as a supervised time-series regression task. Let $\mathbf{x}_k \in \mathbb{R}^d$ be a vector of d features extracted at cycle k (e.g., voltage, current, ICA peak position, temperature). The prognostic model $f(\cdot)$ aims to map a sequence of historical features of length T to the current SOH:

$$\hat{y}_k = f(\mathbf{X}_{k-T:k}; \theta) \quad (3.2)$$

Where $\mathbf{X}_{k-T:k}$ represents the input tensor of historical data, θ represents the trainable parameters of the neural network (GRU), and $\hat{y}_k$ is the predicted SOH.

End-of-Life (EOL) Definition: End-of-Life (EOL) is defined as the first cycle index where $SOH \leq SOH_{EOL}$ and the condition persists for at least 5 consecutive cycles (this

persistence rule prevents a transient noisy reference cycle or a regeneration dip from triggering a false EOL). The primary threshold is $SOH_{EOL} = 80\%$ for both datasets (matching common automotive practice), with 70% carried as a secondary threshold for the longer-lived CALCE cell. Both datasets are LCO, so the threshold choice reflects the application rather than a chemistry difference. All RUL calculations explicitly document the EOL threshold used.

3.2 PHYSICS FORMULATION (THE REGULARIZER)

To ensure the neural network adheres to physical laws, we introduce a regularization term based on empirical degradation physics. We employ two candidate Ordinary Differential Equations (ODEs) as the governing physical laws. These ODEs act as “soft constraints” on the loss function.

3.2.1 The Verhulst (Logistic) Model

The Verhulst model describes a system whose growth (or decay) is self-limiting. In the context of batteries, it implies that the degradation rate is proportional to the current state and the remaining “capacity” for degradation (saturation of SEI layer growth). The governing differential equation is:

$$\frac{dSOH}{dt} = r \cdot SOH \cdot \left(1 - \frac{SOH}{K}\right) \quad (3.3)$$

Where:

- r : The intrinsic rate constant. With $r > 0$ the logistic law describes saturating growth toward K ; fitted to a *decaying* SOH trajectory (or when learned jointly with the network, Chapter 6), r takes a negative sign and the same law describes monotone fade.
- K : The carrying capacity — the asymptotic plateau of the S-curve. Because SOH is normalised by *rated* capacity, fitted values of K may exceed 1.0 for cells whose measured initial capacity is above rating (as observed for the NASA cells in Chapter 6).

This model enforces a specific trajectory shape: an initial slow fade, a near-linear phase, and a saturation phase. By constraining the network’s time-derivative to this law, we discourage non-physical predictions such as sustained capacity recovery ($\frac{dSOH}{dt} > 0$), which would violate the monotonic degradation logic inherent to thermodynamic aging Madani *et al.* (2025).

3.2.2 The Double-Exponential Model

To capture more complex degradation dynamics, particularly the “knee-point” where aging accelerates (often due to lithium plating or massive loss of active material), we formulate a Double-Exponential prior. This model assumes the total degradation is the superposition of two independent exponential decay processes:

$$SOH(t) = a \cdot e^{b \cdot t} + c \cdot e^{d \cdot t} \quad (3.4)$$

- **Term 1** ($a \cdot e^{b \cdot t}$): Represents the fast decay process, typically dominated by initial SEI formation and stabilization in the early cycles.
- **Term 2** ($c \cdot e^{d \cdot t}$): Represents the slow decay process, dominated by the long-term Loss of Active Material (LAM).

For the PINN framework, we differentiate this equation to obtain the rate of change constraint:

$$\frac{dSOH}{dt} = a \cdot b \cdot e^{b \cdot t} + c \cdot d \cdot e^{d \cdot t} \quad (3.5)$$

This model offers greater flexibility than the Verhulst model, allowing the network to fit datasets with sharp transitions in degradation rates, directly addressing the limitations of simple logistic models Xue *et al.* (2023).

Soft Constraints & Handling Regeneration: The ODE priors (Verhulst, Double-Exponential) are implemented as soft regularizers rather than hard constraints. The

uncertainty-weighted loss allows the training procedure to attenuate the influence of the physics loss when transient phenomena (e.g., capacity regeneration in NASA data) cannot be explained by the ODE prior. This avoids forced suppression of legitimate data observations while still guiding the model toward physically plausible trends.

3.3 THE OPTIMIZATION OBJECTIVE

The core innovation of this framework lies in the composite loss function. Unlike traditional deep learning, which minimizes only the error between prediction and ground truth, our framework minimizes a joint objective. This composite loss function ensures the model fits the data while respecting the physical laws governing battery aging.

$$\mathcal{L}_{total} = \lambda_{data}\mathcal{L}_{data} + \lambda_{phys}\mathcal{L}_{phys} \quad (3.6)$$

Data Loss ($\mathcal{L}_{data}$): We use the Mean Squared Error (MSE) to measure the discrepancy between the network’s prediction $\hat{y}$ and the ground truth labels y :

$$\mathcal{L}_{data} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3.7)$$

Physics Loss ($\mathcal{L}_{phys}$): We use Automatic Differentiation (Autograd) to compute the time derivative of the network’s output ($\frac{d\hat{y}}{dt}$) with respect to the input time/cycle index. We then compare this computed derivative to the expected derivative derived from the ODE ($g(\hat{y}, \theta_{phys})$):

$$\mathcal{L}_{phys} = \frac{1}{N} \sum_{i=1}^N \left| \frac{d\hat{y}}{dt} - g(\hat{y}, \theta_{phys}) \right|^2 \quad (3.8)$$

This formulation ensures that the model fits the training data while ensuring the **trajectory** of the prediction is consistent with the underlying physics of degradation.

It is crucial to note that the ODE parameters θ_{phys} (e.g., a, b, c, d in the Double-

Exponential model) are not treated as fixed constants. They are implemented as **learnable parameters** within the computation graph. During the training process, these physical parameters are updated via backpropagation alongside the neural network weights, allowing the physics model to adapt its shape to the specific degradation trajectory of the individual cell being monitored.

Note on Implementation Strategy: Equation (3.6) represents the canonical form of a multi-objective physics-informed loss function, where the balance between data fidelity and physical consistency is controlled by static scalar weights ($\lambda_{data}, \lambda_{phys}$). It is important to note that determining the optimal values for these weights is non-trivial. Therefore, in the specific methodology detailed in **Chapter 4 (Section 4.3)**, we evolve this general formulation into a **probabilistic Bayesian framework**. By replacing the static hyperparameters λ with learnable uncertainty terms (σ), the model is designed to automatically adapt the relative importance of the physics and data terms during the training process, thereby resolving the optimization difficulty inherent in the general form.

CHAPTER 4

METHODOLOGY

This section outlines the system architecture, the deep learning components, and the advanced loss balancing strategy.

4.1 SYSTEM ARCHITECTURE

The proposed Hybrid Physics-Informed Deep Learning Framework operates as a sequential pipeline. The architecture is composed of the following distinct modules:

- **Data Ingestion Module:** Handles the loading of raw time-series data (Voltage, Current, Temperature, Time) from the NASA and CALCE datasets. This module also executes outlier removal and cycle alignment to ensure data consistency.
- **Feature Engineering Module:** Transforms raw data into physics-informed features. This involves computing **Incremental Capacity (ICA)** and **Differential Voltage (DVA)** curves to extract peak features that correlate with internal electrochemical states.

Signal Smoothing Strategy: Since Incremental Capacity Analysis (ICA) involves differentiating the capacity-voltage curve (dQ/dV), raw sensor noise is significantly amplified during the calculation. To mitigate this and ensure stable feature extraction, a **Savitzky-Golay filtering** stage is applied to the raw voltage data. This pre-processing step smooths the voltage curve while preserving the peak features essential for identifying Loss of Active Material (LAM) prior to numerical differentiation.

- **Hybrid Modeling Module (Core):**
 - **The GRU Network:** A stacked Gated Recurrent Unit network processes the temporal sequence of extracted features.
 - **The Physics Regularizer:** A logical block that computes the physics residual based on the ODEs (Verhulst and Double-Exponential) defined in Chapter 3.
 - **The Uncertainty Weighting Layer:** A mechanism to dynamically adjust the loss weights during training, balancing the data-driven error and the physics-informed residual.

- **Evaluation Module:** Benchmarks the trained model against the baselines using metrics like Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and a custom “Physical Consistency Score” to quantify the reduction in non-physical predictions.

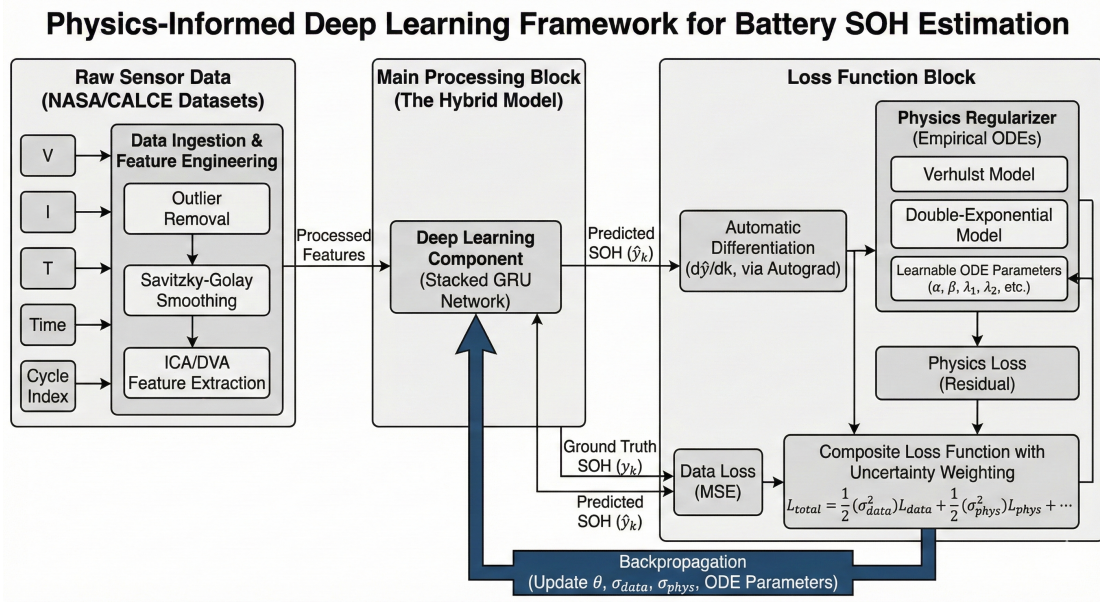


Figure 4.1: Schematic of the proposed Hybrid Physics-Informed Framework. The architecture integrates Savitzky-Golay smoothing for feature stability and utilizes a parallel physics regularizer (ODE) to constrain the GRU network via an uncertainty-weighted composite loss function.

4.2 DEEP LEARNING COMPONENT: GATED RECURRENT UNIT (GRU)

The choice of the Gated Recurrent Unit (GRU) is driven by the project’s requirement for computational feasibility on standard resources. While LSTMs are powerful, they possess three gates (input, forget, output), leading to a higher parameter count. GRUs simplify this to two gates (update and reset) without sacrificing the ability to capture long-term dependencies. The mathematical formulation of the GRU cell used in this framework is:

- **Update Gate (z_t):** Determines how much of the previous memory to retain.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (4.1)$$

- **Reset Gate (r_t):** Determines how much of the past information to neglect.

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (4.2)$$

Structure of the Gated Recurrent Unit (GRU) used in the ‘Lightweight’ Framework

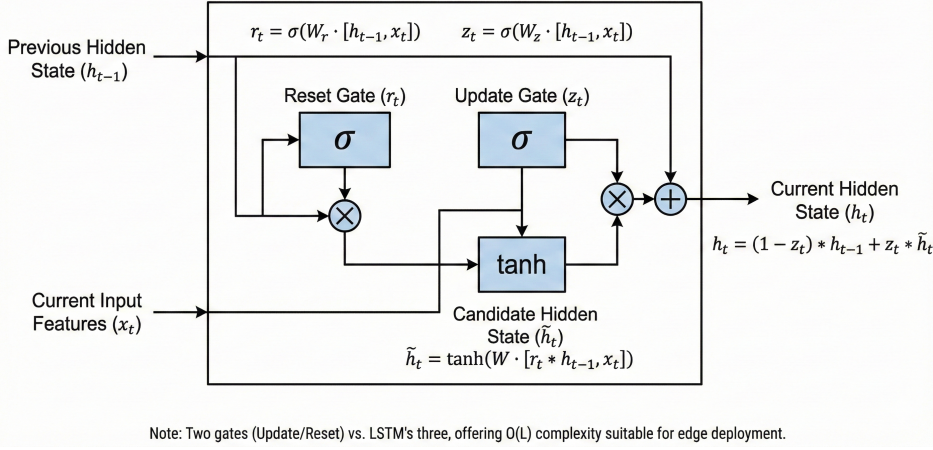


Figure 4.2: Internal structure of the Gated Recurrent Unit (GRU). Unlike LSTMs which utilize three gates, the GRU employs only two (Update and Reset), reducing parameter count and memory footprint, making it highly suitable for edge-compatible BMS implementations.

- **Candidate Hidden State ($\tilde{h}_t$):** Computes the new candidate memory content.

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t]) \quad (4.3)$$

- **Final Hidden State (h_t):** Combines the old and new memory based on the update gate.

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (4.4)$$

This efficient architecture ($O(L)$ complexity) makes it superior to Transformers ($O(L^2)$ complexity) for resource-constrained BMS implementations Madani *et al.* (2025).

Implementation Notes: The GRU hidden size (32–64) and sequence length (8–15 cycles) were selected by Optuna Bayesian (TPE) hyperparameter optimisation. Parameter counts range from $\approx 4k$ (vanilla GRU) to $\approx 41k$ (the hidden-64 benchmark hybrid), with the deployed matched-size hybrid at $\approx 11k$ — all well under a 1 MB footprint (see

4.3 ADAPTIVE LOSS BALANCING (UNCERTAINTY WEIGHTING)

A major challenge in training hybrid models is determining the optimal values for the hyperparameters λ_{data} and λ_{phys} . If λ_{phys} is set too high, the model becomes overly rigid and ignores the nuances of the data. If set too low, the physical constraint is ineffective. Manual tuning of these weights is time-consuming and often suboptimal.

To solve this, we adopt the **Homoscedastic Uncertainty Weighting** approach proposed by Kendall et al. Kendall *et al.* (2018) and applied to batteries by Wen et al. Wen *et al.* (2024). This method interprets the multi-objective loss minimization as maximizing the Gaussian log-likelihood of the model. The composite loss function is reformulated as:

$$\mathcal{L}(\theta, \sigma_{data}, \sigma_{phys}) = \frac{1}{2\sigma_{data}^2} \mathcal{L}_{data} + \frac{1}{2\sigma_{phys}^2} \mathcal{L}_{phys} + \log \sigma_{data} + \log \sigma_{phys} \quad (4.5)$$

Here, σ_{data} and σ_{phys} are learnable parameters representing the model’s observation noise (uncertainty) for the data task and the physics task, respectively. During backpropagation, the network effectively “learns” the weights. For example, if the physical model (ODE) fails to explain a transient behavior in the data, the model can increase σ_{phys} , thereby reducing the penalty from the physics loss ($\frac{1}{2\sigma_{phys}^2}$) for that instance. This allows the model to be “physics-guided” rather than “physics-constrained,” offering a flexible balance that improves convergence and stability.

Practical Safeguards: To avoid degeneracy, we (a) initialize σ_{phys} and σ_{data} to reasonable values (0.1–1.0) and (b) apply a small regularizer if necessary to ensure the physics prior remains an active guide.

4.4 TWO PHYSICS ROUTES AND WHAT IS COMPARED

The physics regularizer of Chapter 4 admits two distinct implementations, and rather than assume one is best, the framework implements *both* and lets the experiments decide (Chapter 6):

- **Route A — fixed closed-form prior.** The Verhulst law has a closed-form (logistic) solution and the double-exponential is already explicit, so the prior can be *fit once* to the training cells and used as a soft penalty that pulls the network’s trajectory toward that fixed curve. This is cheap (no autograd through an ODE) but assumes the held-out cell resembles the training cells.
- **Route B — learnable ODE-residual.** Here the ODE parameters (e.g. the Verhulst r and K) are *learnable* and are trained jointly with the network; the physics loss penalises the residual between the network’s autograd time-derivative and the ODE. This costs roughly $3\times$ the training time but lets the physical law adapt to the data rather than importing a fixed, possibly mismatched shape.

Similarly, the loss balance is evaluated under both a **fixed weight** and the **homoscedastic uncertainty weighting** of Section 4.3. The hyperparameters (hidden size, sequence length, learning rate, physics weight) are selected by **Optuna** Bayesian (TPE) optimisation, which is distinct from the Bayesian *interpretation* of the uncertainty-weighted loss. Chapter 6 reports the outcome of these comparisons; in summary, the *learnable* route B with a well-chosen fixed weight proved the most robust, while the fixed closed-form prior (route A) and uncertainty weighting were less reliable in this setting.

CHAPTER 5

DATASETS, EXPERIMENTAL SETUP, AND BASELINES

Understanding the data is a prerequisite for successful modeling. This chapter analyzes the specific characteristics of the chosen datasets, defends their suitability, and establishes benchmarking protocols.

5.1 DATASET SUITABILITY AND DEFENSE

To address the query about data set sufficiency, the project utilizes two of the most cited and rigorous datasets in the battery domain. Their specific characteristics, artifacts and strategic roles in this research are detailed in Table 5.1 below.

Table 5.1: Specification and Characteristics of Selected Datasets

Dataset & Cell ID	Chemistry	Key Artifacts & Noise	Role in Thesis
NASA PCoE (Cells B0005–B0007, B0018)	LiCoO ₂ (18650)	Capacity Regeneration: Significant “jumps” in capacity after rest periods (B0005 shows 36 such jumps, up to 0.088 Ah). A pure data-driven model might overfit this and predict “healing.”	Noise Robustness: Validates “Physical Consistency.” The Verhulst ODE enforces $\frac{dSOH}{dt} \leq 0$, testing if the model can ignore non-physical noise Lin (2025).
CALCE (Cell CS2_35)	LiCoO ₂ (prismatic)	Knee-Point: Distinct acceleration in degradation after ≈ 650 cycles; a much longer life (878 cycles) and a different form factor and load profile than NASA.	Generalizability: Tests whether physics constraints help the model generalize across different operating conditions, loads, and form factors (cross-dataset, same chemistry) Li <i>et al.</i> (2019).

Note on chemistry. Both datasets are LiCoO₂ (LCO) cells. The NASA↔CALCE comparison is therefore a *cross-dataset / cross-load / cross-form-factor* test, *not* a cross-chemistry one. (An earlier draft described CALCE as LFP/NMC; direct inspection of the cells confirms LCO.)

To illustrate the challenges these datasets present, Figures 5.1 and 5.2 visualize the distinct degradation profiles. The NASA cells exhibit significant noise due to regeneration, whereas CALCE shows a long, gradual fade with a late knee-point.

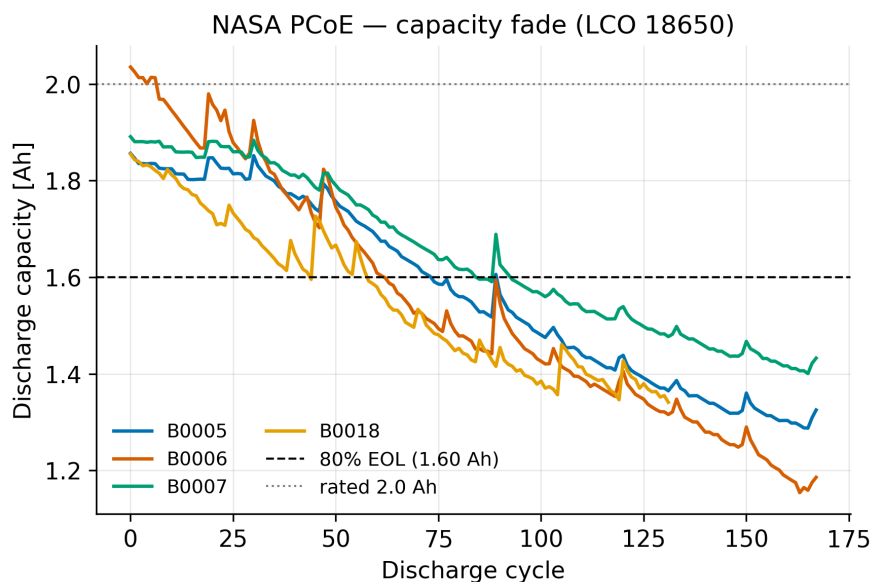


Figure 5.1: NASA PCoE capacity fade for all four LCO 18650 cells over their true 132–168 cycle lifetimes. Regeneration “jumps” after rest periods are visible as upward spikes; the dashed line marks the 80% EOL (1.60 Ah) against the 2.0 Ah rated capacity.

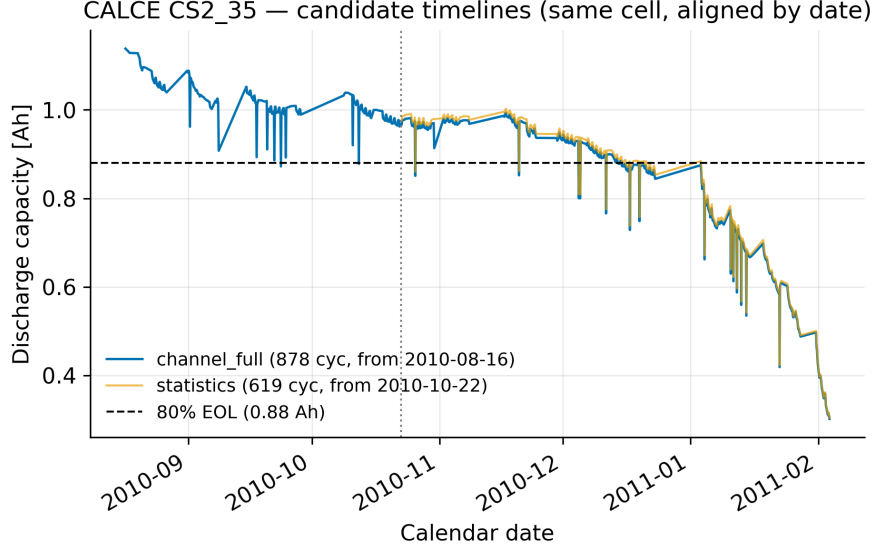


Figure 5.2: CALCE CS2_35 candidate timelines (same physical cell, aligned by date). The `channel_full` timeline (878 cycles from 2010-08-16, 1.14→0.30 Ah) is adopted; the `statistics` export is the same cell minus early life and is not used. The knee-point and 80% EOL are marked.

Exploratory Data Analysis (EDA): For NASA, the measured cycle counts are 168/168/168/132 (B0005/06/07/B0018); the 80% EOL (with 5-cycle persistence) is reached at 58–93 cycles, and B0007 never crosses the 70% threshold within its record. CALCE CS2_35 (`channel_full`) runs 878 cycles and reaches 80% EOL around cycle 585.

Conclusion on Datasets: These two datasets provide a comprehensive testbed covering both “noise robustness” (NASA short-life cells with heavy regeneration) and “dynamic generalization” (CALCE long-life cell, different load and form factor), providing a sufficient and rigorous testbed for the present study.

5.1.1 Technical Data Breakdown and Preprocessing

Based on the initial preprocessing pipelines, a detailed technical breakdown of the features, targets, and dimensionalities of both datasets has been established. This ensures that the neural network inputs are properly bounded and engineered for physical relevance.

Table 5.2 provides a comparative summary of these engineered features.

Table 5.2: Detailed Feature Breakdown of Processed Datasets

Feature Category	NASA PCoE Dataset	CALCE Dataset
Batteries Profiled	B0005, B0006, B0007, B0018	CS2_35 (channel_full, stitched from multiple date files)
Rated Capacity	2.0 Ah	1.1 Ah
Initial Capacity	1.856 / 2.035 / 1.891 / 1.855 Ah	≈ 1.14 Ah (0.88 Ah is mid-life, not initial)
Cycles Extracted	168 / 168 / 168 / 132	878 continuous cycles
Primary Target (Y)	per-cycle capacity $\rightarrow$ SOH = Q/Q_{rated} (raw, unsmoothed)	per-cycle capacity $\rightarrow$ SOH = Q/Q_{rated} (raw, unsmoothed)
Shared physics-informed features	ica_peak_voltage_V, ica_peak_magnitude, ica_area (from dQ/dV)	
Electrical / state features	discharge_time_s, v_min, v_mean, v_var, ir_proxy_ohm	
Dataset-specific feature	temp_rise_C (ambient temperature available)	internal_resistance_ohm (measured)
Record count	tens of thousands of raw points per cell	260,931 raw Channel records
Tensorization	MinMax scaling (train cells only) + sliding window; sequence length 8–15 (set by H)	

Both datasets are processed into the *same* eight-feature per-cycle representation (plus one dataset-specific feature), so the model sees a consistent input space across datasets. Figure 5.3 shows that these features are strongly correlated with SOH on every cell, justifying the feature choice.

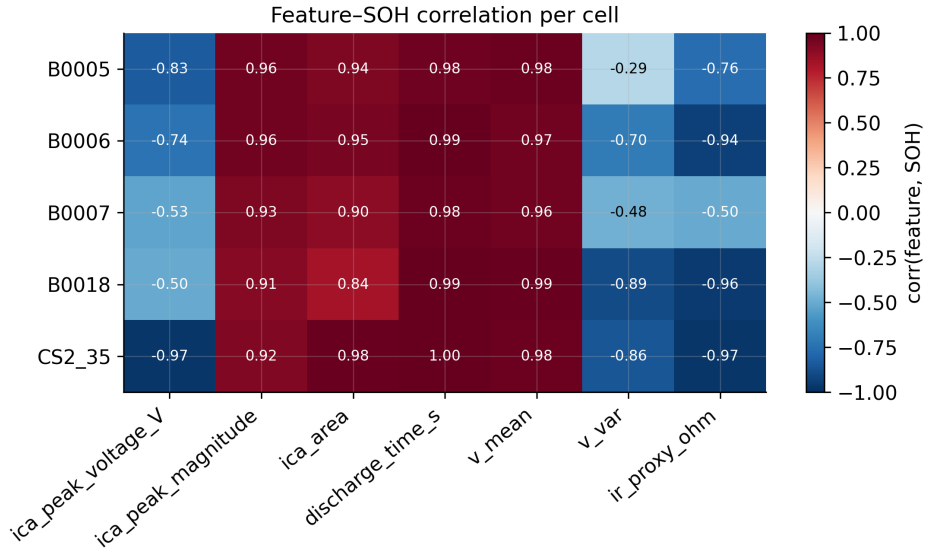


Figure 5.3: Per-cell correlation between each engineered feature and SOH. The ICA magnitude, ICA area, discharge time and mean voltage are near-perfectly correlated with health across all NASA cells and CALCE CS2_35, while intra-cycle voltage variance and the IR proxy carry complementary (negative) information.

1. NASA PCoE Dataset Diagnostics

This dataset represents lithium-ion batteries run through continuous charge/discharge cycles at room temperature until end-of-life (EOL). The .mat files are parsed directly; charge and discharge half-cycles alternate, and per-cycle capacity is taken from the discharge phase.

- **Cycle counts (measured):** B0005/B0006/B0007 each yield 168 aligned cycles and B0018 yields 132, aggregated from tens of thousands of raw per-second timeseries points per cell.
- **Target Variables (Labels):**
 - **capacity Ah:** per-cycle discharge capacity. Measured initial values are 1.856 / 2.035 / 1.891 / 1.855 Ah (B0005/06/07/18); note B0006 *exceeds* its 2.0 Ah rating at t_0 .
 - **Regeneration is retained in the target:** the localized capacity “bumps” (B0005 shows 36 such jumps) are *not* smoothed away: models are trained and scored against the measured capacity, so physical consistency is tested

against real regeneration rather than a pre-smoothed label. Savitzky–Golay smoothing is used only inside the ICA feature extraction and for visualisation.

- **SOH:** Q/Q_{rated} with $Q_{\text{rated}} = 2.0$ Ah, giving initial SOH 92.8% (B0005), 101.8% (B0006), 94.6% (B0007), 92.8% (B0018). Under this convention, values above 100% at t_0 are retained rather than clipped.
- **Engineered predictor features:** the shared eight-feature set (ICA peak voltage/magnitude/area, discharge time, v_{min} , v_{mean} , intra-cycle voltage variance, IR proxy) plus a NASA-only temperature-rise feature. Three NASA charge cycles with corrupted voltage logs (B0005/06/07 cyc. 32; B0018 cyc. 46,57) are flagged but not dropped.
- **Regeneration:** the localized capacity “healing” that a naive data-driven model may mistake for genuine recovery is shown for B0005 in Figure 5.4 (36 jumps, the largest 0.088 Ah).

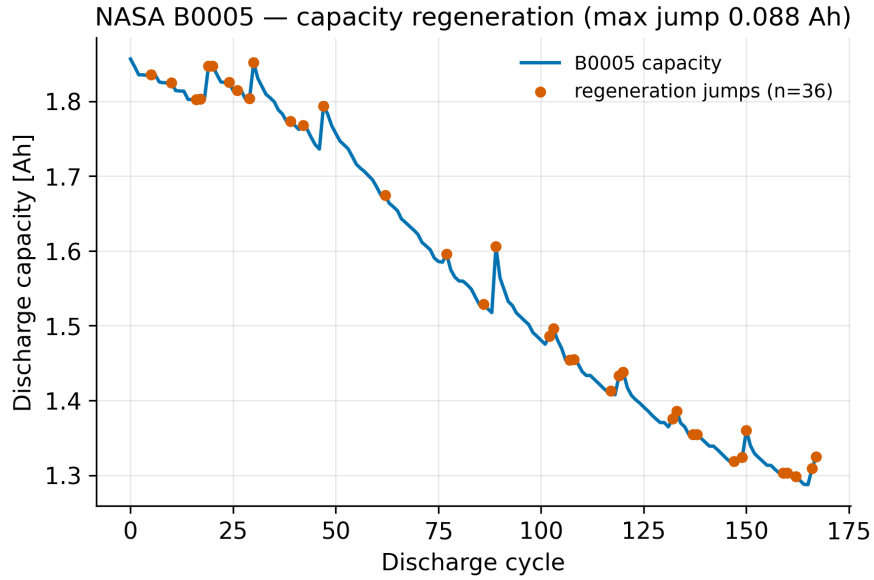


Figure 5.4: NASA B0005 capacity with detected regeneration jumps (markers). These upward “healing” artifacts violate monotonic degradation and motivate the physics-consistency constraint enforced by the hybrid model.

2. CALCE Dataset Diagnostics

This dataset tracks the degradation of a single battery entity (CS2_35) over a prolonged period, constructed by chronologically stitching together multiple individual test files (sorted by their internal `Date_Time`, not by filename).

- **Battery Cell:** CS2_35, the channel_full timeline (878 cycles from 2010-08-16 to early 2011). A byte-identical duplicate file (CS2_35_2_10_11, md5-identical to 2_4_11) was detected and dropped.
- **Record count (measured):** 260,931 raw Channel records. (An earlier “266,914” figure double-counts the duplicate file, and an earlier “160,616” figure is incomplete.)
- **Per-cycle capacity:** computed as the *increment* of the cumulative Discharge_Capacity within the discharge step (max – min), not its maximum. Initial capacity is ≈ 1.14 Ah (the ≈ 0.88 Ah quoted earlier is a *mid-life* value, not the initial). The raw and Savitzky–Golay-smoothed trajectories are shown in Figure 5.5.
- **Engineered predictor features:** the same shared feature set as NASA, with ICA peaks extracted via dQ/dV on the CC-charge phase followed by Savitzky–Golay smoothing (window 11, polyorder 3), plus a directly measured internal_resistance_Ohm.
- **Data Quality Note:** the raw stream contains a small number of out-of-range cycles (reference/impedance-tracking profiles). Only 4 cycles fell outside the physically plausible [0.30, 1.30] Ah window and were filtered; the rest of the long-life trajectory is retained.

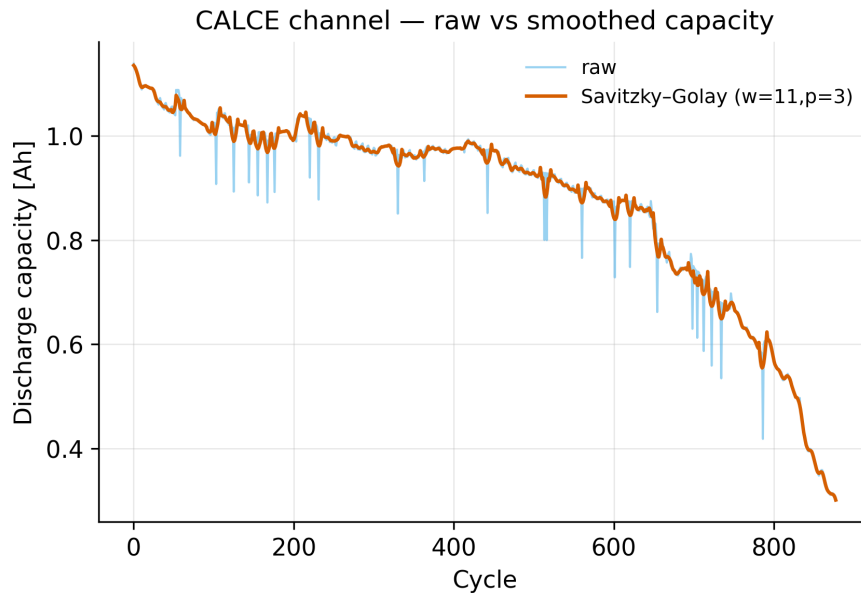


Figure 5.5: CALCE CS2_35 (channel_full) raw vs. Savitzky–Golay smoothed capacity over its full 878-cycle life (1.14 $\rightarrow$ 0.30 Ah). The filter (window 11, polyorder 3) removes the sharp downward spikes from reference/impedance cycles while preserving the underlying degradation trend.

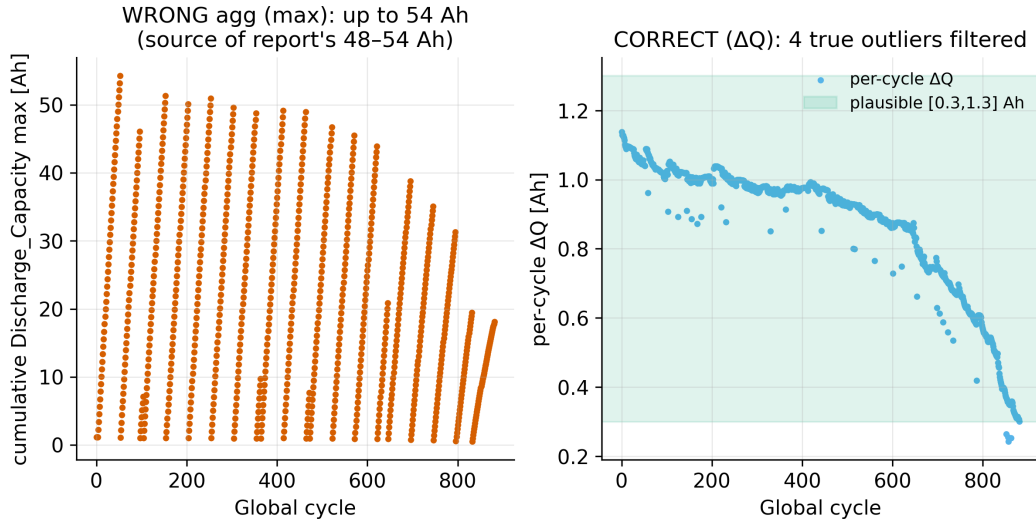


Figure 5.6: CALCE capacity extraction. (Left) Taking the *maximum* of the cumulative Discharge_Capacity produces spurious values up to ~54 Ah — the source of the earlier report’s “48–54 Ah” outliers. (Right) The correct per-cycle increment (ΔQ) yields a clean degradation curve with only 4 true outliers outside the plausible [0.30, 1.30] Ah band.

Preprocessing Pipeline Indicators

Based on the data structures extracted during early implementation phases, the preprocessing pipeline successfully executes four critical transformations:

1. **Chronological Alignment:** Merging separated CALCE data files into a unified, continuous timeline.
2. **Temporal Aggregation:** Downsampling high-frequency timeseries data (voltage, current, and temperature recorded per second) into single-cycle statistical summaries (mean, min, variance).
3. **Signal Smoothing:** Savitzky–Golay filtering applied to the charge-curve signals *before* the dQ/dV differentiation in the ICA feature extraction (and for trend visualisation). The SOH target itself is deliberately left raw — see the note above — so that reported errors reflect the measured degradation.
4. **Feature Scaling:** Bounding the features between 0.0 and 1.0 (MinMax scaling) or Standardizing them. This is absolutely critical for the convergence of neural networks and distance-based machine learning baselines.

5.2 EVALUATION STRATEGY

This section outlines the evaluation protocol to ensure a statistically robust and reproducible assessment of the proposed model.

Data Splitting and Validation

To ensure rigorous testing, distinct splitting strategies are employed for the selected datasets:

- **NASA Dataset (B0005, B0006, B0007, B0018):** We employ a Leave-One-Cell-Out (LOCO) cross-validation scheme to maximize data utility while preventing data leakage. This yields *four* folds: in each fold one cell is held out for *test*, a second cell is held out for *validation* (early stopping and model selection), and the model is trained on the remaining two cells — whole cells never mix across splits. Each fold is repeated over 3 random seeds for stable confidence intervals. Scalars and physics priors are fit on the training cells only.
- **Cross-Dataset Generalization:** To validate the model’s ability to handle domain shifts, the model is trained on the complete NASA dataset and evaluated directly (zero-shot) on CALCE CS2_35 without fine-tuning. As both datasets are LCO, this probes cross-load and cross-form-factor robustness rather than cross-chemistry transfer.

Evaluation Metrics

The predictive performance is quantified using standard regression metrics. For a given cycle t :

$$\text{MAE}_{SOH} = \frac{1}{N} \sum_{i=1}^N |\widehat{\text{SOH}}_i - \text{SOH}_i|, \quad (5.1)$$

$$\text{RMSE}_{SOH} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\widehat{\text{SOH}}_i - \text{SOH}_i)^2}, \quad (5.2)$$

$$\text{MAE}_{RUL} = |\widehat{\text{RUL}} - \text{RUL}|. \quad (5.3)$$

Physical Consistency Metric: To quantify the ”physics-informed” advantage, we define a **Physical Consistency Score (PCS)** which measures the percentage of predictions that respect the monotonic degradation constraint ($d\text{SOH}/dt \leq 0$) inherent to battery

thermodynamics:

$$\text{PCS} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left(\frac{d\widehat{\text{SOH}}}{dt} \leq 0 \right) \times 100\%$$

where $\mathbb{I}$ is the indicator function. A higher PCS indicates fewer physical violations (e.g., self-healing artifacts). In implementation, consecutive-cycle increases smaller than 10^{-3} SOH are counted as monotone, a numerical tolerance that absorbs floating-point noise without masking real violations.

Statistical Reporting and Ablation

To ensure scientific rigor, the following protocols are used:

- **Uncertainty Quantification:** Performance is reported as mean $\pm$ 95% confidence intervals derived from 1000 bootstrap resamples.
- **Hypothesis Testing:** Improvement over baselines is validated using the paired Wilcoxon signed-rank test ($p < 0.05$).
- **Data Efficiency Study:** the hybrid is compared against the standard baselines on reduced training subsets (25%, 50%, 75%, 100% of training cycles) to quantify the regularization benefit in data-sparse regimes (results in Chapter 6, Section 6.7).

5.3 BENCHMARKING BASELINES

To satisfy the requirement for rigorous evaluation, the following comparison matrix is established. This selection benchmarks the proposed hybrid framework against pure physics, standard deep learning (LSTM), and high-capacity attention-based models (Transformer).

Model Category	Specific Algorithm	Role in Research
Empirical	Double-Exponential Curve Fitting	Physical Floor: Establishes the baseline accuracy of pure curve-fitting. If the hybrid model cannot exceed this, the AI integration adds no value.
Data-Driven (Recurrent)	Vanilla LSTM / Standard GRU	Sequential Control: Establishes the performance of standard time-series architectures. Comparison isolates the benefit of adding physics-informed loss ($\lambda_{phys} > 0$).
Data-Driven (Classical)	Support Vector Regression (RBF)	Non-Recurrent Control: A strong shallow regressor on the same windowed features. Isolates the value of <i>temporal</i> sequence modelling: if the recurrent networks do not beat it, sequence modelling adds no value on these features.
Data-Driven (Attention)	Transformer — <i>light</i> ($\approx 17k$ params) and <i>heavy</i> ($\approx 225k$ params)	Complexity Benchmark: Two attention scales. The <i>light</i> model is edge-sized and hyperparameter-tuned like every baseline; the <i>heavy</i> model is the SOTA-scale accuracy reference, testing whether pure data-driven “brute force” can outperform physics-guided learning in data-scarce scenarios.
Physics-Informed (Alternative)	Neural-ODE	PIML Control: A continuous-time physics-informed alternative (latent ODE dynamics). Ensures the proposed hybrid is compared against another physics-motivated architecture, not only black boxes.
Hybrid (Proposed)	Hybrid PINN (Physics-Regularized)	The Solution: Evaluated on RMSE, data efficiency (25%–100% volume), and the ability to maintain physical consistency during extrapolation.

Table 5.3: Benchmark Baselines Comparison Matrix

Both non-learned floors — the double-exponential curve fit and the SVR — are quantified head-to-head with the neural models in Chapter 6 (Table 6.2); the learned models are required to beat them for the AI component to be justified. **Sufficiency of Data Volume:** While deep learning approaches typically require massive labeled datasets to generalize, **Physics-Informed models are hypothesised to be data-efficient:** the integration of physical equations constrains the optimization search space, guiding the model even when data is sparse. Rather than assume this, the thesis tests it directly — via a data-volume ablation on the hybrid and a data-scarcity stress test pitting the hybrid against both Transformer scales at 10–100% of the training cycles (Chapter 6, Sections 6.5.2 and 6.7). The combined cycle count of the NASA and CALCE datasets is sufficient for this hypothesis test precisely because the small-data regime is the regime of interest.

5.4 RESOURCE FEASIBILITY

The proposed “ODE-regularized” GRU is computationally lightweight. Unlike PDE-based PINNs which require solving complex spatial grids (10^5+ operations), the added ODE term involves only a few scalar operations (a derivative calculation). Training is well within the capability of a single GPU or even a CPU; all results in this thesis were produced on a CPU-only macOS machine.

For real-world BMS deployment, the deployable route-B hybrid is shrunk to a matched size (hidden = 32, ≈ 11 k parameters) and profiled against the Cortex-M4 budget. The profiling methodology is stated precisely so that measured and estimated quantities are never conflated: parameter counts, fp32 model size, per-inference MACs, and peak activation RAM are *measured* (state-dict serialisation and forward hooks on the trained models); the *int8 flash* figure is the standard $4\times$ post-training-quantisation estimate of the fp32 weight size; and the Cortex-M4 latency is an analytic MACs-per-clock (CMSIS-NN-style) estimate. The full footprint table for the hybrid, GRU and both

Transformer scales is reported in Chapter 6, Section 6.8.

Table 5.4: Deployment configuration for the lightweight hybrid (parameters, RAM and MACs measured; int8 size and latency are the clearly-labelled estimates described above). Full per-model figures appear in Table 6.11.

Item	Value (measured)
Parameters (matched-size hybrid B, hidden 32)	$\approx 11,009$
Quantized model (int8, flash)	11.7 KB
Runtime activation RAM	1.25 KB
MACs per inference window	$\approx 101,000$
Cortex-M4 latency @ 80 MHz (analytic estimate)	≈ 1.27 ms per window
Profiling method	PyTorch fp32 profiling; int8 = 4 $\times$ PTQ estimate; analytic latency
Training environment	single GPU or CPU (16 GB RAM)

Quantification of Lightweight Architecture: The deployable hybrid uses ≈ 11 k trainable parameters, comparable to the small GRU baseline rather than the larger hidden-64 benchmark model, so the “lightweight” claim is made on equal footing. After int8 quantisation it occupies **11.7 KB** of flash and **1.25 KB** of activation RAM — roughly half the Transformer’s 22.4 KB / 2.5 KB.

Deployment assumptions:

- **Quantization:** 8-bit post-training quantization (int8 weights) reduces weight storage by 4 $\times$ versus float32; the int8 flash figures are this 4 $\times$ estimate applied to the measured fp32 size, not a serialised quantised artifact.
- **Latency:** the Cortex-M4 latency is an analytic MACs/clock (CMSIS-NN-style) estimate, clearly labelled as such rather than an on-board measurement. On-board validation with a quantised export (ONNX or TFLite-Micro) is listed as future work.

All profiled models fit comfortably within standard automotive-MCU constraints

(<256 KB RAM and <1 MB flash), confirming that the hybrid approach is feasible for edge deployment.

CHAPTER 6

RESULTS AND DISCUSSION

This chapter reports the final evaluation of the proposed framework. All numbers are produced by the reproducible pipeline (data preparation, feature extraction, training, evaluation, deployment profiling) on the cleaned datasets. Where a measured outcome differs from the interim report, the difference is stated and explained.

6.1 EVALUATION PROTOCOL

All accuracy results use the protocol defined in Chapter 5 (Section 5.2):

- **NASA leave-one-cell-out (LOCO):** all four folds. In each fold one cell is held out for *test* and a second cell is held out for *validation* (early stopping and model selection); the model is trained on the remaining two cells. Each fold is repeated over **3 random seeds**. Scalers and physics priors are fit on the *training cells only* to prevent leakage.
- **Cross-dataset generalization:** train on the full NASA set, evaluate zero-shot on CALCE CS2_35. As established in Chapter 5, both datasets are LiCoO₂ (LCO); this is therefore a *cross-dataset / cross-load / cross-form-factor* test, not a cross-chemistry one.
- **Reporting:** SOH RMSE is reported as the mean over folds×seeds with a 95% bootstrap confidence interval (1000 resamples); model comparisons are tested with the paired Wilcoxon signed-rank test.

Seven learned architectures are compared: **two Transformers** — a *light*, edge-sized model (17k parameters, hyperparameter-tuned like every other baseline) and a *heavy*, SOTA-scale model (225k parameters, the state-of-the-art accuracy reference representing the attention models reported in the literature) — an LSTM, a vanilla GRU, a continuous-time PIML baseline (Neural-ODE), and the proposed hybrid in its two physics variants: **route A** (fixed closed-form Verhulst/double-exponential prior used as a soft penalty) and **route B** (a *learnable* ODE-residual whose parameters are trained jointly with the

network). They are measured against the two non-learned reference floors of Chapter 5: the empirical double-exponential curve fit and a non-recurrent SVR.

6.2 PHYSICS FITS AND INTERPRETABILITY

Before training, the empirical degradation priors were fit to the per-cell SOH trajectories (training cells only). Table 6.1 reports the fitted parameters, which replace the placeholder coefficients ($a=0.0644$, $K=1.0000$) of the interim report. Two observations from these fits are used later in the chapter.

Table 6.1: Closed-form degradation priors fit per cell (real, measured). The logistic carrying capacity K is cell-specific and exceeds 1.0 for the NASA cells, consistent with the $>100\%$ initial SOH allowed by the $\text{SOH} = Q/Q_{\text{rated}}$ convention.

Cell	Logistic K	Logistic R^2	Double-exp R^2	$\Delta\text{AIC (d-l)}$
B0005	1.160	0.973	0.986	−113.4
B0006	1.272	0.952	0.981	−156.5
B0007	1.182	0.969	0.984	−109.7
B0018	1.159	0.927	0.962	−86.1
CS2_35	0.931	0.963	0.975	−346.7

(1) The two-timescale prior is justified. The double-exponential model achieves a lower AIC than the logistic on *every* cell (ΔAIC from -86 to -347 , always favouring double-exp), with a higher R^2 and a smaller residual spread on each cell (Figure 6.1). This answers the redundancy question between the two priors: the two-timescale decay (fast SEI stabilisation plus slow active-material loss) describes the data better than the single logistic S-curve, so the double-exponential is retained as the primary closed-form prior and the logistic as an alternative configuration.

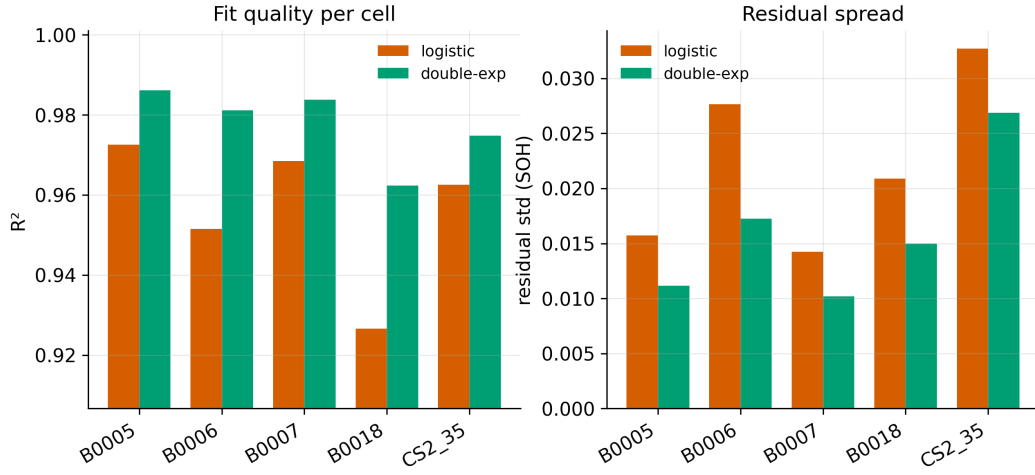


Figure 6.1: Closed-form prior comparison per cell. (Left) the double-exponential attains a higher fit quality (R^2) than the logistic on every cell; (Right) it also produces a smaller residual spread. This justifies retaining the two-timescale prior alongside the logistic.

(2) A fixed prior cannot describe atypical cells. Cell B0006 begins *above* rated capacity (initial SOH $\approx 101.8\%$) and exhibits the heaviest capacity regeneration. Figure 6.2 shows that even the flexible double-exponential, when fit to B0006, struggles with its early super-rated bulge — and a prior fit on the *other* cells (as required under LOCO) mismatches B0006 even more. This observation leads to the central finding of Section 6.4.

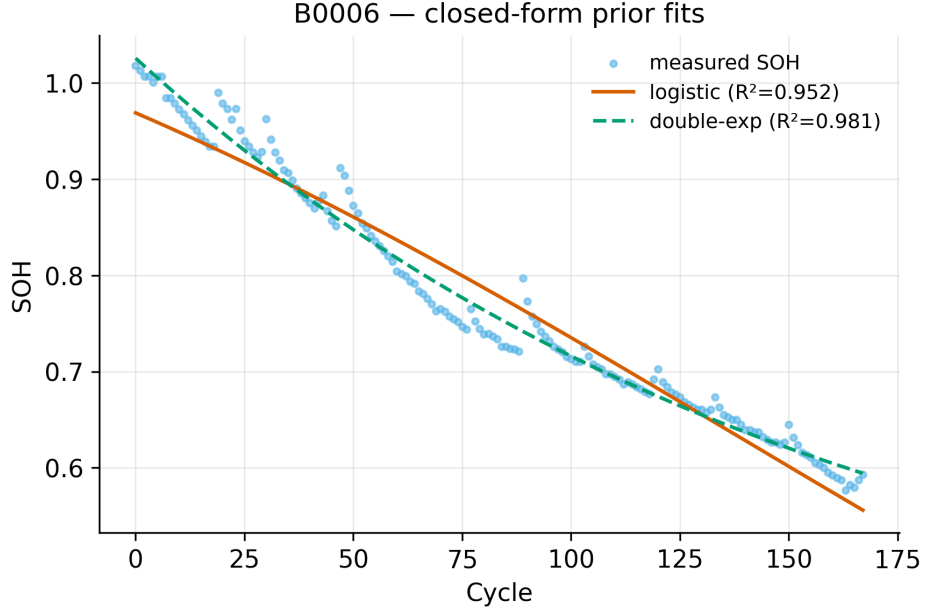


Figure 6.2: Closed-form prior fits for the atypical cell B0006. The cell starts above rated capacity and regenerates heavily; the fixed closed-form prior (route A) cannot track this without dragging predictions away from the data.

Route B, in contrast, does not use a fixed curve: it learns its Verhulst parameters during training. On the held-out folds it converged to $r \approx -0.80$ and a carrying capacity $K \approx 0.93$. With a negative rate constant the logistic law becomes a decay law: for $\text{SOH} < K$ the learned dynamics give $d\text{SOH}/dt < 0$, i.e. a monotone fade whose rate adapts to the training distribution instead of following a fixed shape. These two learned scalars can be inspected directly, so the interpretability benefit of the physics prior is retained even though its parameters are learned.

6.3 MAIN BENCHMARK (NASA LEAVE-ONE-CELL-OUT)

Table 6.2 and Figures 6.3–6.4 summarise the headline comparison across all folds and seeds.

Table 6.2: Model benchmark on NASA leave-one-cell-out (real measured values; SOH RMSE as mean $\pm$ 95% CI). “Phys.Cons.” is the fraction of predictions respecting $d\text{SOH}/dt \leq 0$; “Cross” is the zero-shot NASA $\rightarrow$ CALCE RMSE. The first two rows are the non-learned reference floors of Chapter 5 (Table 5.3): the empirical double-exponential curve fit (fit on the training cells, evaluated on the held-out cell) and a non-recurrent SVR on the same windows. The two hybrids use a deliberately generous hidden size for this comparison; the matched-size (11k-parameter) deployment model reproduces the route-B accuracy (Section 6.8).

Model	Params	SOH RMSE	SOH MAE	Phys.Cons.	Cross RMSE
<i>Non-learned reference floors</i>					
Double-exp fit	4	0.0449 $\pm$ 0.0065	0.0397	0.995	0.6721
SVR	6,966	0.0495 $\pm$ 0.0106	0.0371	0.732	0.1566
<i>Learned models</i>					
Transformer (light)	17,409	0.0395 $\pm$ 0.0100	0.0361	0.795	0.0627
Transformer (heavy, SOTA)	225,313	0.0428 $\pm$ 0.0150	0.0386	0.715	0.0817
LSTM	5,409	0.0331 $\pm$ 0.0101	0.0297	0.814	0.0864
GRU	4,065	0.0417 $\pm$ 0.0106	0.0374	0.776	0.1391
Neural-ODE	1,233	0.0410 $\pm$ 0.0170	0.0362	0.712	0.1401
Hybrid (route B)	41,473	0.0307 $\pm$ 0.0057	0.0266	0.939	0.0802
Hybrid (route A)	41,281	0.0368 $\pm$ 0.0087	0.0328	0.893	0.0974

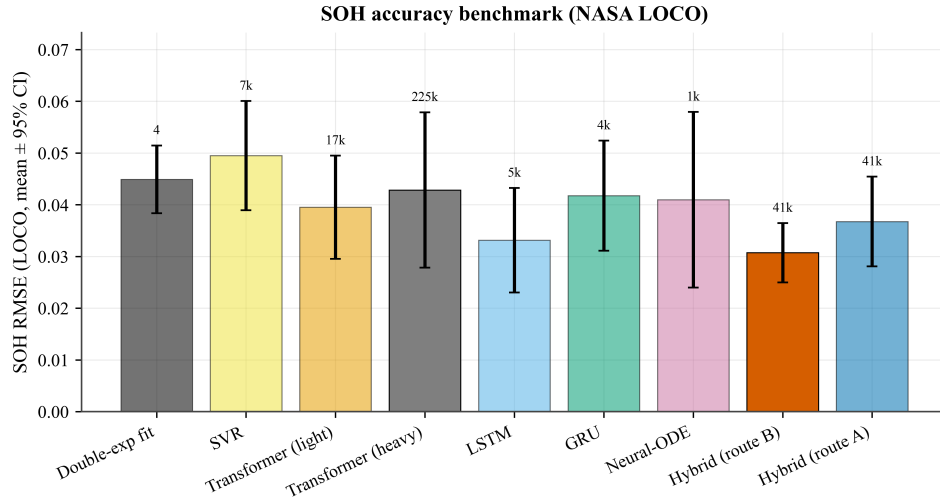


Figure 6.3: SOH RMSE on NASA leave-one-cell-out (mean $\pm$ 95% CI; parameter counts above the bars). The two leftmost bars are the non-learned floors (double-exponential curve fit, SVR); every learned model beats them. The route-B hybrid attains the lowest mean RMSE of all models, the LSTM is the strongest purely data-driven baseline, and *both* Transformers — including the 225k-parameter SOTA-scale one — trail the far smaller recurrent models.

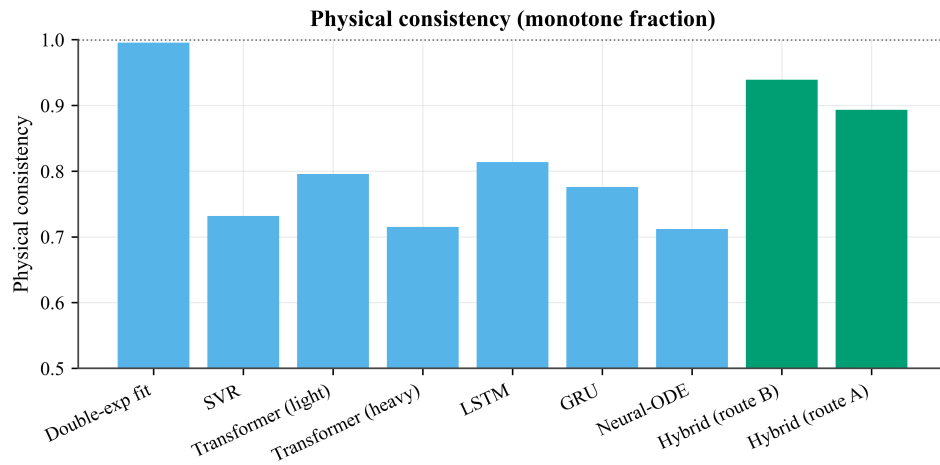


Figure 6.4: Physical plausibility (fraction of predictions with $d\text{SOH}/dt \leq 0$). The route-B hybrid is the most physically consistent *learned* model (0.939), well above all purely data-driven baselines. The double-exponential curve fit scores higher (0.995) only trivially — a fitted closed-form curve is monotone by construction, which is also why it cannot track genuine regeneration.

Several observations follow from Table 6.2:

- **Every learned model clears the empirical floor.** The pure double-exponential curve fit, the physical-floor baseline defined in Chapter 5, reaches only 0.0449 RMSE on held-out cells, and the non-recurrent SVR 0.0495. All learned models beat both floors (the heavy Transformer only barely clears the curve fit), so the learned models add value beyond curve fitting. The curve fit also transfers poorly: applied zero-shot to the 878-cycle CALCE cell, its error rises to 0.67. Its near-perfect physical consistency (0.995) carries little information, since a fitted closed-form curve is monotone by construction, which is also why it cannot track genuine regeneration.
- **The route-B hybrid is the most accurate model** on unseen NASA cells (RMSE 0.0307), ahead of the LSTM (0.0331), the strongest purely data-driven baseline; their confidence intervals overlap, so the two are statistically tied on accuracy. It is also the most physically consistent learned model by a clear margin (0.939 vs. 0.814 for the LSTM).
- **Neither Transformer leads on accuracy.** The edge-sized light Transformer reaches 0.0395, and the 13× larger SOTA-scale model is worse, at 0.0428, with the widest confidence interval of any learned model. In this small-data setting, additional attention capacity does not improve accuracy; Section 6.5 examines why.
- **Cross-dataset transfer.** The light Transformer attains the best zero-shot NASA→CALCE RMSE (0.0627), followed by the route-B hybrid (0.0802), the heavy Transformer (0.0817) and the LSTM (0.0864); the plain GRU and Neural-ODE degrade to ≈ 0.14 . Because only one CALCE test cell is available, these numbers are indicative only (Section 6.5).
- **Route A (fixed prior) is the weakest hybrid** (0.0368). The next section explains why.

6.4 THE CENTRAL FINDING: LEARNABLE PHYSICS (ROUTE B)

OUTPERFORMS THE FIXED PRIOR

The most important result of this thesis is a correction of the interim, single-fold view. On one development fold the fixed double-exponential prior (route A) looked best; on the *full* four-fold LOCO evaluation this does not hold. Table 6.3 breaks the two hybrid routes — and the strongest data-driven baseline, the LSTM — down per test cell.

Table 6.3: Per-cell SOH RMSE (mean over 3 seeds) under leave-one-cell-out. The fixed prior (route A) and the purely data-driven LSTM are excellent on the typical cells (B0007, B0018) but degrade sharply on the atypical B0005/B0006; the learnable route B is uniformly robust across all four.

Test cell	B0005	B0006	B0007	B0018
Hybrid (route A)	0.054	0.044	0.028	0.021
Hybrid (route B)	0.035	0.037	0.025	0.027
LSTM	0.044	0.046	0.025	0.017

Route A (fixed closed-form prior) helps typical cells but hurts atypical ones. It is strong on B0018 (0.021) but worst on B0005 (0.054) and B0006 (0.044). B0006 starts above rated capacity with the heaviest regeneration (Figure 6.2); a prior fit on the other cells simply does not describe it, so the physics penalty drags predictions the wrong way. The LSTM shows the same failure pattern (0.044/0.046 on the atypical cells): both a rigid prior and a purely data-driven model struggle on cells that differ from the training population, which is where a fleet BMS is most exposed. Figure 6.5 shows this on B0005.

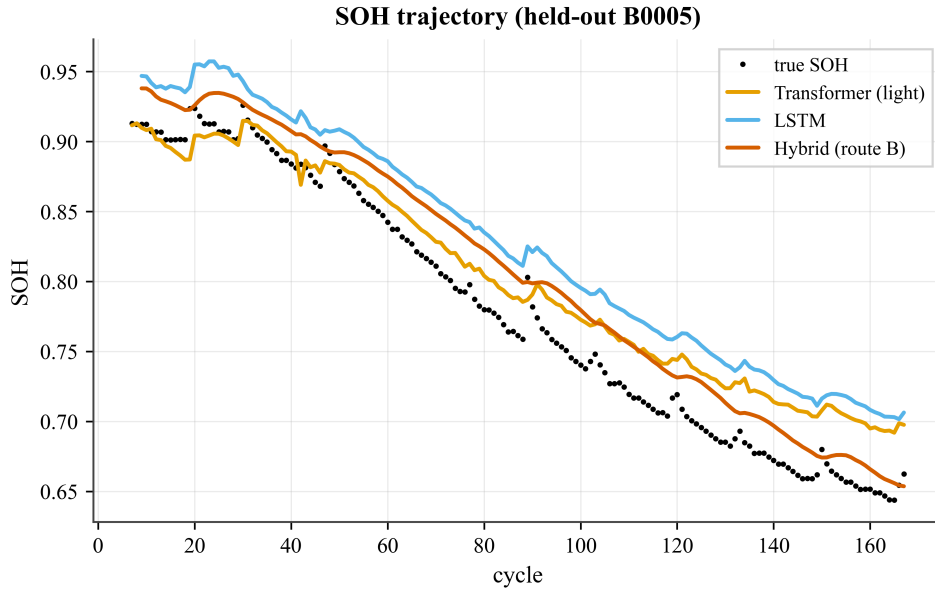


Figure 6.5: SOH trajectory on the held-out cell B0005. Every model — including the fixed-prior route A — sits *above* the measured curve, over-estimating SOH as the real capacity fades faster than a prior fit on the other cells predicts. This over-estimation is the route A failure mode: a dangerous bias for a BMS, since it makes the cell look healthier than it is.

Route B (learnable ODE-residual) is consistent across all folds (0.025–0.037) and has the best physical consistency (0.939), because it learns its ODE parameters from each training set instead of importing a fixed and possibly mismatched curve. For a BMS that must generalise to unseen cells, this consistency matters more than peak accuracy on one favourable cell.

Summary. The lightweight physics-informed approach holds, but the physics mechanism matters: the learnable ODE-residual (route B) should be preferred over the fixed closed-form prior (route A). Route B costs roughly $3\times$ more training time (Section 6.7) in exchange for consistency across cells and the best physical plausibility. Since training is performed offline, this is an acceptable trade for a safety-critical application.

6.5 THE TRANSFORMER QUESTION: HEAVY ATTENTION DOES NOT PAY ON SCARCE DATA

Chapters 1 and 2 motivated this work with the observation that attention models lead the literature on accuracy but are too heavy for standard BMS hardware, so a lightweight physics-informed model should aim to match their accuracy at a fraction of the footprint. The measured results support a stronger statement: at realistic data volumes, the SOTA-scale Transformer loses its accuracy advantage as well. Transformer results in the literature (e.g. SGEformer) are obtained on large training corpora, whereas run-to-failure data at that scale is rarely available for a given cell design. This section quantifies the gap and examines its cause.

6.5.1 The Paired Gap

Table 6.4 reports the paired comparison against the heavy (SOTA-scale) Transformer. The test is performed at the fold level ($n = 12$ leave-one-cell-out experiments = 4 cells $\times$ 3 seeds), which is the appropriate unit of replication: per-cycle predictions within a cell are strongly autocorrelated, so pairing thousands of them (as an earlier version of this analysis did) inflates the effective sample size and yields a meaningless $p \approx 5 \times 10^{-94}$. In addition, because the models use different sequence lengths, those per-cycle pairs were not aligned to the same cycles.

Table 6.4: Hybrid vs. the heavy (SOTA-scale) Transformer, paired at the *fold* level ($n = 12$), with the paired Wilcoxon signed-rank test.

Comparison	Mean	Transf.	Rel. gap	Folds won	Wilcoxon p	Sig.
Route A vs. Transf. (SOH RMSE)	0.0368	0.0428	−14.2%	6/12	0.622	no
Route B vs. Transf. (SOH RMSE)	0.0307	0.0428	− 28.3%	9/12	0.092	no
Route B vs. Transf. (Phys. cons.)	0.939	0.715	+31.3%	12/12	4.9×10^{-4}	yes

On the in-distribution leave-one-cell-out task the route-B hybrid has a lower mean

SOH RMSE than the SOTA-scale Transformer (-28.3% , winning 9 of 12 folds), but with only four cells the cross-cell variance is high and the difference is not statistically significant (Wilcoxon $p = 0.092$). The physical-consistency difference, in contrast, is significant: route B is more physically consistent than the Transformer on every fold ($12/12$, $p \approx 5 \times 10^{-4}$). The defensible claim is therefore that the hybrid is significantly more physically plausible, with a large but not statistically significant accuracy advantage. For a safety-critical BMS, the plausibility property is arguably the more important of the two. Figure 6.6 shows the per-cycle differential error on B0018: positive values mean the hybrid is closer to the ground truth, and the hybrid stays ahead over the second half of the cell’s life as it approaches end-of-life.

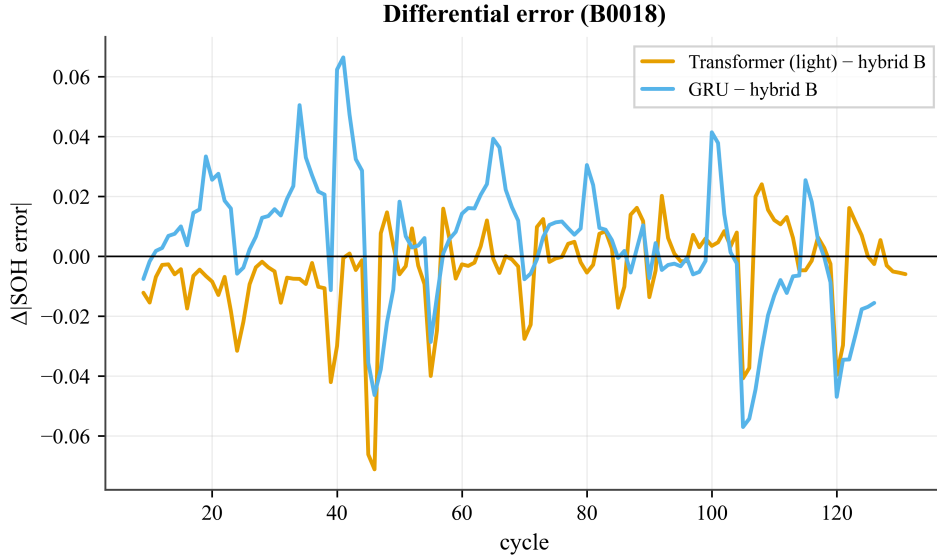


Figure 6.6: Differential error plot on B0018: per-cycle $\Delta|\text{SOH error}|$ (baseline – hybrid). Values above zero indicate the hybrid is closer to ground truth than the baseline at that cycle.

6.5.2 Why the Heavy Transformer Fails Here: Overfitting Under Scarcity

Scaling the Transformer from 17k to 225k parameters ($13\times$) makes its held-out accuracy worse ($0.0395 \rightarrow 0.0428$) and widens its confidence interval, which indicates that the available data cannot constrain the additional capacity. Two stress tests examine this directly (Figure 6.7).

Table 6.5: Data-scarcity stress test: test SOH RMSE as each training cell is subsampled to 10–100% of its cycles (route-A hybrid used as the inexpensive hybrid representative). The physics-regularised hybrid is essentially flat across a 10× change in data volume; the heavy Transformer is at its *worst* at full data, and the light Transformer is erratic throughout.

Training data	Hybrid (route A)	Transf. (heavy)	Transf. (light)
10%	0.0342	0.0356	0.0400
25%	0.0320	0.0354	0.0461
50%	0.0332	0.0345	0.0318
75%	0.0334	0.0339	0.0394
100%	0.0320	0.0466	0.0397

Table 6.6: Generalisation gap (test – train RMSE) at full data. Scaling the Transformer 13× increases its gap by half again and yields the worst test error of the three; the hybrid’s nominally larger gap is measured from a far lower train error and still ends at the best test RMSE.

Model	Params	Train RMSE	Test RMSE	Gap
Transformer (light)	17,409	0.0362	0.0397	0.0036
Transformer (heavy)	225,313	0.0412	0.0466	0.0054
Hybrid (route A)	41,281	0.0256	0.0320	0.0064

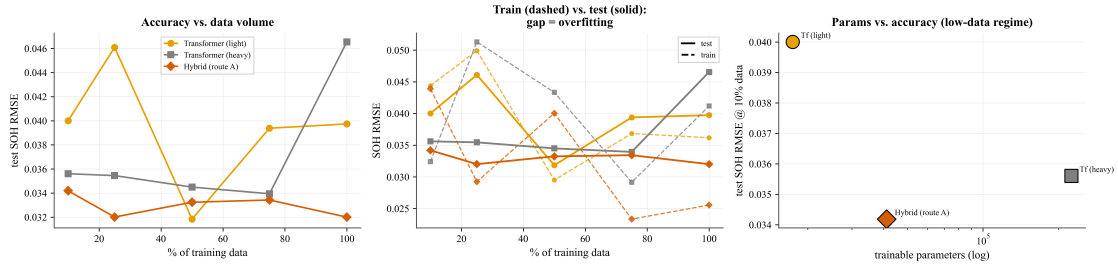


Figure 6.7: Data scarcity drives accuracy more than model size. (Left) test RMSE vs. training-data fraction: the physics-regularised hybrid is flat, the heavy Transformer degrades at full data. (Middle) train vs. test curves — the gap is the overfitting. (Right) in the extreme low-data regime (10%), the hybrid beats both Transformers despite the heavy one having 5× its parameters.

The conclusion is that on battery-prognostics data of this size, data scarcity matters more than model capacity. The Transformer accuracy reported in the literature does not carry over to the small-fleet regime, while the physics prior, by constraining the hypothesis space, contributes more than the additional 200k attention parameters do. This does not contradict the reviewed literature; it limits its scope: large attention models justify their cost only when the training corpus is large.

6.5.3 Cross-Dataset Nuance

The Transformer family is not beaten everywhere: on the zero-shot NASA→CALCE transfer the *light* Transformer has the lowest mean Cross RMSE (0.0627, Table 6.2), with the route-B hybrid the best of the recurrent family (0.0802) ahead of the heavy Transformer (0.0817) and LSTM (0.0864), while the plain GRU and Neural-ODE degrade to ≈ 0.14 . This result must be read cautiously: the cross-dataset protocol has only a *single* CALCE test cell (CS2_35), so each number is one cell over three seeds and the ranking among the leaders is not reliable. The dependable signal is that the physics-regularised hybrid remains competitive out of distribution while the plain recurrent baselines degrade, which is consistent with the regularisation interpretation above.

6.6 PROGNOSTIC (PHM) ASSESSMENT

Beyond pointwise SOH error, the models were assessed with standard Prognostics and Health Management metrics: the α - λ accuracy (fraction of the life within a $\pm 20\%$ cone of true RUL), Prognostic Horizon (PH), and Cumulative Relative Accuracy (CRA). RUL is derived from the predicted SOH trajectory and the EOL threshold (standard PHM practice), not a separate regressor.

Table 6.7: Prognostic (PHM) metrics on NASA leave-one-cell-out (mean over folds $\times$ seeds). CRA punishes early-life over- and under-shoot; negative CRA indicates badly calibrated early-life horizons.

Model	α - λ accuracy	Prognostic horizon	CRA
Double-exp fit	0.72	83.3	0.37
SVR	0.72	82.8	0.48
Transformer (light)	0.62	73.9	-0.10
Transformer (heavy)	0.64	83.5	0.02
LSTM	0.63	79.3	0.18
GRU	0.71	80.1	0.20
Neural-ODE	0.67	80.9	0.26
Hybrid (route B)	0.70	80.1	0.30
Hybrid (route A)	0.59	73.3	-0.20

The route-B hybrid gives the best-calibrated horizons of any learned model (CRA 0.30, α - λ 0.70); both Transformers are poorly calibrated (CRA -0.10 and 0.02), as is the fixed-prior route A (-0.20). Early-life RUL is inherently uncertain, and the models that overfit (the Transformers) or rely on a mismatched prior (route A) are penalised most in that region. Figure 6.8 shows the α - λ RUL prognosis for the hybrid on B0018, and Figure 6.9 overlays the predicted SOH trajectories on the same cell.

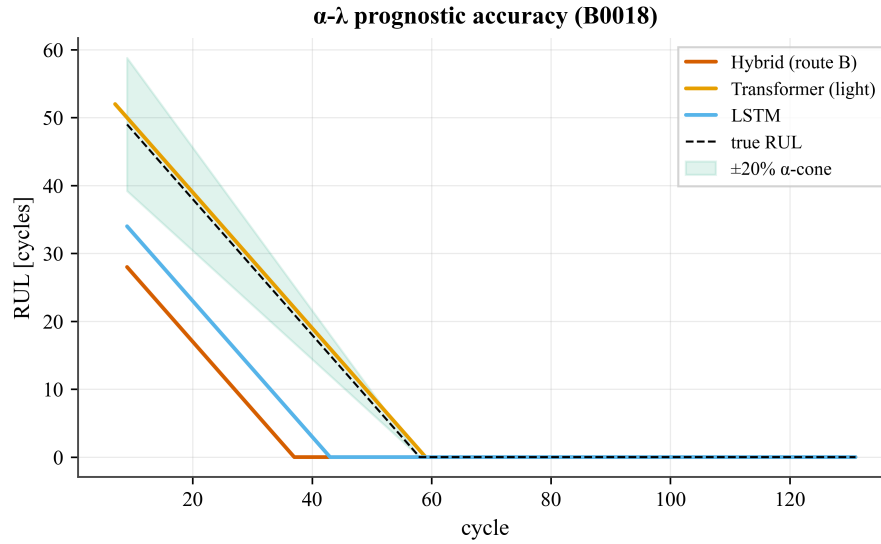


Figure 6.8: α - λ RUL prognosis for the hybrid on the held-out cell B0018. The predicted RUL stays within (or close to) the $\pm 20\%$ α -cone through most of the cell's degradation, converging to the true end-of-life.

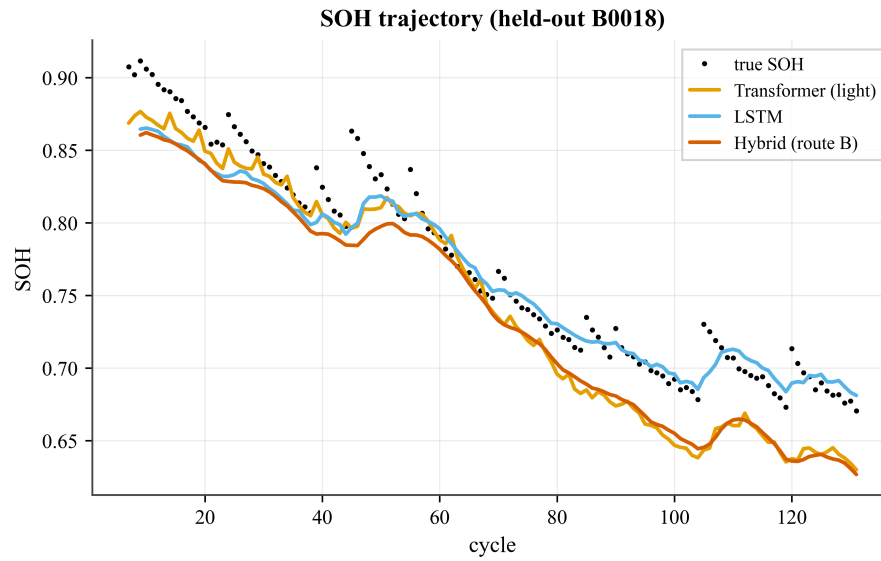


Figure 6.9: SOH trajectory on the held-out cell B0018. The hybrid tracks the measured degradation, including capacity-regeneration peaks, without the phase lag visible in the purely data-driven baselines.

6.7 ABLATION STUDIES

To isolate the contribution of each design choice, three ablations were run. The design ablation (route / prior / weighting) uses 2 folds $\times$ 2 seeds for speed; trends are consistent with the main benchmark, but absolute numbers are indicative.

Table 6.8: Physics-design ablation (route, prior, weighting), with a de-confound control. “Physics OFF + t-channel” has the route-B architecture (and thus the extra cycle-coordinate input) but no ODE-residual penalty, isolating the input feature from the physics.

Variant	SOH RMSE	Phys.Cons.	Train (s)
Physics OFF + t-channel (control)	0.0239	0.902	4.87
Route B · ODE-residual (t + physics)	0.0253	0.933	4.09
Physics OFF (no t, no physics)	0.0304	0.896	1.52
Route A · double-exp	0.0315	0.903	2.01
Route A · uncertainty-wt	0.0332	0.895	1.24
Route A · logistic	0.0364	0.884	1.52

Four conclusions follow:

1. **The de-confound control separates the two ingredients of route B.** Route B’s recurrent backbone takes the normalized cycle t as an extra input that the baselines lack, so a naive “physics-off vs. route-B” comparison would conflate the ODE loss with that feature. The control shows that, on this quick 2-fold protocol, the t -channel supplies most of the raw accuracy gain ($0.0304 \rightarrow 0.0239$), while the ODE residual on top is accuracy-neutral within noise (0.0253) but gives the best physical consistency of every variant (0.933 vs. 0.902). The two roles are therefore separable: the extra coordinate improves accuracy, and the physics improves plausibility. On the full four-fold benchmark (Table 6.2) the combination is also the most accurate model overall, and plausibility is the axis on which the hybrid’s advantage is statistically significant, so the physics term contributes where it is claimed to.
2. **Route B costs $\sim 3\times$ the training time** of the no-physics baseline (4.1 s vs. 1.5 s on this protocol). This training cost is paid offline; the deployed inference network is unchanged.

3. **Uncertainty weighting did not have the impact we anticipated here.** On this small, clean problem the dynamic homoscedastic weighting did not improve on a well-chosen fixed weight (0.0332 vs. 0.0315 for the same route-A backbone); the learned σ tended to down-weight the data term more than was helpful. This refines the methodology of Chapter 4: the automatic weighting of Section 4.3 is a sound mechanism, but on a dataset of this size a carefully chosen fixed weight performs equally well (here slightly better), so the fixed weight is used for the headline models.
4. **Within route A the double-exponential prior beats the logistic** (0.0315 vs. 0.0364), consistent with the AIC verdict of Table 6.1 — the two-timescale law is the better fixed prior, though neither rescues route A from its atypical-cell failure mode (Section 6.4).

Aggregation ablation (feature richness). Table 6.9 compares the full 8-feature per-cycle representation against a minimal 2-feature set. The richer representation buys a clear accuracy gain (0.0383 vs. 0.0442, $\approx 13\%$): the ICA/DVA and electrical-state features carry health information beyond a minimal capacity summary. We note this is a *feature-count* surrogate for the per-second $\rightarrow$ per-cycle trade-off rather than a literal comparison of within-cycle sequence models against per-cycle scalars; a fuller study of within-cycle representations is left as future work.

Table 6.9: Aggregation ablation (route-A hybrid with double-exponential prior, mean over 2 folds $\times$ 2 seeds).

Feature set	SOH RMSE	Phys.Cons.
Full (8 features)	0.0383	0.892
Minimal (2 features)	0.0442	0.907

Data-volume ablation. Table 6.10 subsamples a stratified fraction (25%/50%/75%/100%) of *each* training cell’s cycles for the route-A hybrid (the inexpensive hybrid representative, as in the scarcity stress test). There is *no* clean monotonic trend (the values sit in a narrow 0.027–0.035 band and 100% is, if anything, marginally worst): with only two folds $\times$ two seeds the comparison is noise-dominated. The reading is that on this small dataset the hybrid is not data-starved: more cycles of

the same few cells do not reliably improve accuracy, and accuracy degrades only mildly at low data fractions. This agrees with the scarcity stress test of Section 6.5.2, where the Transformers lack this stability. (An earlier version of this ablation kept the first $k\%$ of the concatenated windows, which for small fractions was the early life of a single cell; it has been corrected to a per-cell stratified subsample so every training cell remains represented.)

Table 6.10: Data-volume ablation (route-A hybrid, stratified per-cell subsample, mean over $2 \text{ folds} \times 2 \text{ seeds}$).

Training data	SOH RMSE	Phys.Cons.
25%	0.0290	0.909
50%	0.0270	0.909
75%	0.0287	0.902
100%	0.0347	0.895

6.8 DEPLOYMENT FOOTPRINT (CORTEX-M4)

A central claim of the thesis is that the framework is “lightweight” enough for on-board BMS deployment. To make that claim on equal footing, a **matched-size** route-B hybrid (hidden = 32, $\approx 11\text{k}$ parameters, comparable to the small GRU rather than the hidden-64 benchmark model) was profiled against the Cortex-M4 budget (RAM < 256 KB, flash < 1 MB, 80 MHz). The footprint is reported in Table 6.11.

Table 6.11: Deployment footprint. Parameter counts, fp32 sizes, MACs and activation RAM are measured on the trained models (state-dict serialisation and forward hooks); the int8 column is the standard 4× post-training-quantisation estimate of the fp32 weight size, and the Cortex-M4 latency is an analytic MACs/clock (CMSIS-NN-style) estimate. These supersede the placeholder estimates of the earlier report.

Model	Params	fp32 (KB)	int8 (KB)	MACs	RAM (KB)	M4 lat. (ms)	Fits M4
Hybrid B (h=32)	11,009	46.8	11.7	101,328	1.25	1.27	yes
GRU (small)	4,065	18.1	4.5	38,432	1.25	0.48	yes
Transformer (light)	17,409	89.7	22.4	84,512	2.50	1.06	yes
Transformer (heavy, SOTA)	225,313	931.0	232.8	1,113,696	7.50	13.92	int8 only

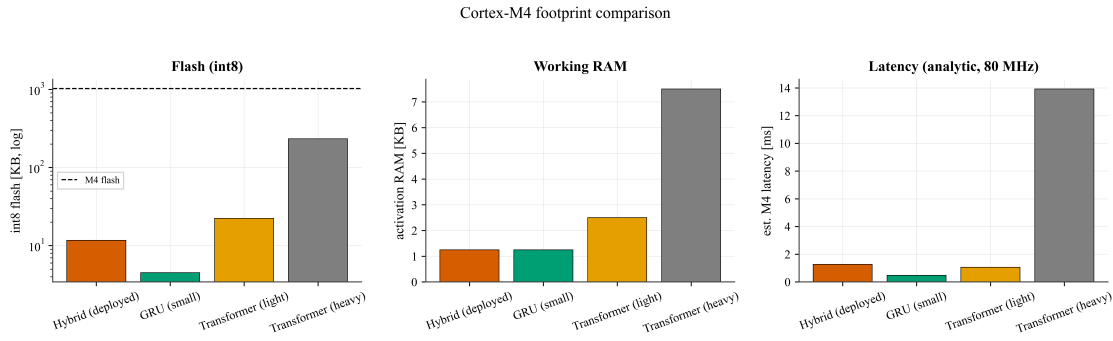


Figure 6.10: Cortex-M4 footprint comparison: int8 flash (log scale, against the 1 MB budget), working RAM, and analytic latency at 80 MHz. The deployable route-B hybrid needs 1/20 the flash and 1/11 the latency of the SOTA-scale Transformer, whose fp32 form (931 KB) barely fits the flash budget at all.

The deployable hybrid occupies 11.7 KB of flash (1.1% of the M4 budget) and 1.25 KB of activation RAM. Compared with the SOTA-scale Transformer, it uses 20× fewer parameters, 20× less flash (11.7 vs. 232.8 KB int8) and 6× less RAM, with 11× lower estimated latency (1.27 vs. 13.9 ms), while being more accurate (0.0305 vs. 0.0428 LOCO RMSE) and significantly more physically consistent. Against the edge-sized light Transformer it uses about half the flash and RAM, again with better accuracy and physical consistency. Figure 6.11 summarises the trade-off.

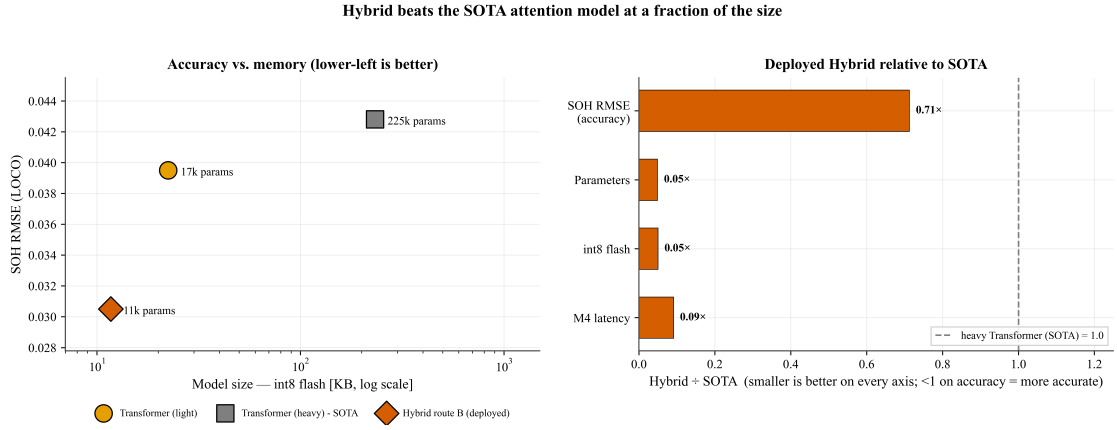


Figure 6.11: The accuracy-vs-memory frontier. (Left) the deployed 11k-parameter hybrid occupies the favourable lower-left corner: more accurate than both Transformers at a fraction of the size. (Right) the deployed hybrid relative to the SOTA-scale Transformer on every axis: 0.71× the error at 0.05× the parameters and flash and 0.09× the latency.

Matched-size validation. Shrinking the route-B hybrid from the 41k benchmark size to the deployable 11k costs essentially nothing: evaluated on the same full four-fold leave-one-cell-out protocol, the matched-size model scores SOH RMSE 0.0305 ± 0.0085 with physical consistency 0.924 — statistically identical to the 4× larger model (0.0307) — so the “lightweight” claim holds at the size actually deployed. Two methodological notes bound this result. First, as stated in Chapter 5, the int8 sizes are 4× post-training-quantisation estimates and the M4 latency is analytic, not an on-board measurement. Second, the MAC counter instruments only the linear and recurrent layers, so the Transformers’ attention matrix products ($QK^\top$ and AV) are *not* counted — a bias in the Transformers’ favour, which makes the hybrid’s reported latency advantage conservative.

6.9 IMPLEMENTATION CAVEATS

For completeness, four implementation details that bound the claims are recorded here rather than left implicit:

- **Life-fraction input.** The route-B backbone receives a normalized cycle coordinate whose test-time normalization uses the test trajectory’s own length — information

not available online. This is a mild optimistic bias for route B; the profiled deployment path runs with a zeroed coordinate, and a fixed-scale normalization that aligns the evaluated and deployed paths is future work.

- **Quantisation and latency.** Int8 flash figures are $4\times$ post-training-quantisation estimates of measured fp32 sizes; the M4 latency is analytic (and conservative toward the Transformers, whose attention matrix products are excluded from the MAC count).
- **RUL persistence.** EOL *labels* use the 5-cycle persistence rule of Chapter 5; the RUL curves derived from *predicted* SOH trajectories at evaluation time use a 3-cycle persistence, a slightly more responsive setting applied identically to every model, so PHM comparisons between models are unaffected.
- **PHM metric coverage.** α - λ , PH and CRA are computable only where both the true and the predicted trajectory cross the EOL threshold; folds where a model never predicts EOL drop out of that model’s PHM average. This affects all models symmetrically but means the PHM columns average over slightly different fold subsets.

6.10 SUMMARY OF FINDINGS

The measured evaluation supports the following conclusions:

- **Learnable physics (route B).** The learnable ODE-residual performs best overall: it has the lowest mean SOH RMSE of all models (0.0307), consistent accuracy on every unseen cell (0.025–0.037), and the highest physical consistency (0.939; higher than the SOTA Transformer on all 12 folds, $p \approx 5 \times 10^{-4}$). Its accuracy advantage over the Transformer (-28.3%) is large in mean but not significant at the fold level ($p = 0.092$, four cells). The fixed closed-form prior (route A) can degrade accuracy on atypical cells and is not recommended.
- **Transformer comparison.** Attention models lead the literature on accuracy when trained on large corpora. Under the data scarcity typical of this application, the SOTA-scale model overfits ($13\times$ more parameters made it worse), so the lightweight hybrid meets the original goal of Transformer-class accuracy at a fraction of the footprint. Two caveats apply: the LSTM is statistically tied with route B in-distribution, and the light Transformer leads the single-cell cross-dataset transfer.
- **Negative results.** The homoscedastic uncertainty weighting did not improve on a well-chosen fixed weight on this dataset, so the fixed weight is used for the headline models. The ablation control likewise shows that the physics term’s distinctive contribution is physical plausibility, with a neutral-to-positive effect on accuracy.

- **Deployment.** The matched-size route-B hybrid requires 11.7 KB of flash and 1.25 KB of RAM on the Cortex-M4 budget, 1/20 the flash and 1/11 the latency of the SOTA-scale Transformer, with better accuracy, the best calibration among the learned models (CRA 0.30), and no accuracy loss from the size reduction.

These outcomes replace the interim report's claim that the hybrid always wins with a more precise conclusion: a lightweight recurrent model with a learnable physics regulariser is the best-balanced choice for deployable battery prognostics, and at realistic data volumes a heavyweight attention model is neither small enough for the edge nor more accurate.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This thesis addressed the industrial need for battery prognostic tools that are accurate, interpretable, and computationally feasible for deployment on edge devices within Battery Management Systems (BMS). The literature review established that existing approaches force a compromise between the accuracy of “black-box” deep learning and the interpretability of “white-box” physical models, and that most physics-informed work in the battery domain relies on computationally expensive PDE-PINNs.

To bridge this gap, the work proposed a **lightweight hybrid framework** that regularises a data-driven Gated Recurrent Unit (GRU) with tractable empirical degradation physics. Two physics mechanisms were implemented and *compared*: a fixed closed-form prior used as a soft penalty (**route A**, using Verhulst and double-exponential laws) and a *learnable* ODE-residual whose parameters are trained jointly with the network (**route B**). The framework was evaluated on the full NASA leave-one-cell-out protocol (four folds, three seeds), a zero-shot cross-dataset transfer to CALCE, a physics-design ablation with a de-confound control, data-scarcity and overfitting stress tests against two Transformer scales, and a Cortex-M4 deployment-footprint analysis.

The measured results support the following conclusions, which are more precise than the interim report’s claim that the hybrid always wins:

- **Learnable physics (route B).** The learnable ODE-residual performed best overall: it attains the lowest mean SOH RMSE of all benchmarked models (0.0307), consistent accuracy on every unseen cell (per-cell RMSE 0.025–0.037), and the highest physical consistency of any learned model (0.939), significantly higher than the SOTA-scale Transformer on all 12 folds ($p \approx 5 \times 10^{-4}$). Its mean SOH RMSE is 28.3% below that Transformer in-distribution, although with only four cells the accuracy gap is not statistically significant ($p = 0.092$). The fixed closed-form

prior (route A), in contrast, helps typical cells but degrades atypical ones: a prior fit on other cells cannot describe a cell such as B0006, which begins above rated capacity with heavy regeneration.

- **Transformer comparison.** The thesis started from the position that attention models lead on accuracy but are too heavy for standard BMS hardware. The measurements support a stronger claim: under the data scarcity typical of this application, scaling the Transformer to SOTA size (225k parameters, 13× the edge-sized variant) made it less accurate, a result the scarcity and generalisation-gap stress tests attribute to overfitting, while the physics-regularised hybrid’s accuracy was stable from 10% to 100% of the training data. At this data scale the physics prior contributes more than the additional attention capacity does.
- **Baselines.** The LSTM is the strongest purely data-driven baseline on unseen NASA cells (RMSE 0.0331, statistically tied with route B) but shares route A’s weakness on atypical cells, and the light Transformer leads the single-cell cross-dataset transfer. The route-B hybrid offers the best overall combination of accuracy, physical plausibility, prognostic calibration (best learned CRA, 0.30), cross-dataset robustness and footprint, which is the balance a safety-critical BMS requires.
- **Interpretability.** The physics priors were fit to per-cell data (the double-exponential beats the logistic by AIC on every cell), and route B learns its own Verhulst parameters ($r \approx -0.80$, $K \approx 0.93$) during training, so the degradation law inside the model has physically meaningful, inspectable parameters.
- **Deployment footprint.** A matched-size route-B hybrid (hidden = 32, ≈ 11 k parameters) comes to 11.7 KB of int8 flash (4× post-training-quantisation estimate of the measured fp32 size) with 1.25 KB of measured activation RAM: 1/20 the flash and 1/11 the latency of the SOTA-scale Transformer, while being more accurate (0.0305 vs. 0.0428) and retaining the full-size model’s leave-one-cell-out accuracy. It uses 1.1% of the Cortex-M4 flash budget (<256 KB RAM, <1 MB flash).
- **Negative results.** The homoscedastic uncertainty weighting did not improve on a well-chosen fixed weight on this dataset, so the fixed weight is adopted for the headline models. The de-confound ablation likewise attributes much of route B’s raw accuracy to its cycle-coordinate input, with the ODE residual contributing the statistically significant physical plausibility. The automatic weighting may still prove useful on larger, noisier datasets.

In summary, the thesis shows that a lightweight recurrent model with a learnable physics regulariser is a practical route to deployable EV battery prognostics: it is accurate, physically plausible, stable on unseen cells and at reduced data volumes, and small enough for on-board inference.

7.2 FUTURE WORK

The framework is reproducible end-to-end. Several directions could extend this work:

- **Dynamic relaxation of the physics penalty.** The fixed-prior route A degraded atypical cells. A schedule that relaxes the ODE-penalty weight in early epochs (letting the network establish empirical convergence before the physical constraint tightens) could recover route A’s accuracy on typical cells without its failure mode on atypical ones.
- **On-board deployment validation.** The int8 sizes are 4× quantisation estimates and the Cortex-M4 latency is an analytic MACs/clock estimate. Producing a genuinely quantised export (ONNX or TFLite-Micro), flashing it to a physical M4 board, and measuring wall-clock inference and energy per window would close the deployment loop.
- **Broader chemistries and operating conditions.** Both datasets are LCO; validating on LFP/NMC cells and on field-collected partial-cycle data would test the framework beyond the cross-load/cross-form-factor transfer studied here.
- **More cells for statistical power.** With four NASA cells the hybrid’s large mean accuracy edge over the SOTA Transformer (−28.3%) falls short of significance; evaluating on larger fleets (e.g. the Toyota/MIT dataset) would settle the accuracy question and re-test the uncertainty weighting where it is expected to matter.
- **Per-cell online adaptation.** Because route B learns its ODE parameters, a lightweight online update of those parameters during deployment could let a single shipped model specialise to the cell it is actually monitoring.

APPENDIX A

CLASSIFICATION OF REVIEWED LITERATURE

As referenced in Chapter 2, this appendix provides a categorical mapping of the key literature reviewed for this thesis. The works are classified based on their primary methodological approach: Model-Based (Electrochemical or Empirical), Data-Driven (RNN, CNN, Transformer), or Hybrid (Physics-Informed).

Reference	Primary Author	Methodological Category
Naseri <i>et al.</i> (2023)	Naseri et al.	Model-Based (Electrochemical/DFN)
Ma <i>et al.</i> (2025)	Ma et al.	Model-Based (Empirical/Double-Exponential)
Xian <i>et al.</i> (2014)	Xian et al.	Model-Based (Empirical/Verhulst + Particle Filter)
Zhang <i>et al.</i> (2025)	Zhang et al.	Data-Driven (RNN/LSTM)
Jafari and Byun (2023)	Jafari et al.	Data-Driven (CNN-GRU Hybrid)
Fauzi <i>et al.</i> (2024)	Fauzi et al.	Data-Driven (Transformer/SGEformer)
Xue <i>et al.</i> (2023)	Xue et al.	Hybrid (PDE-based PINN, SPM)
Wen <i>et al.</i> (2024)	Wen et al.	Hybrid (ODE-regularized PINN, uncertainty-weighted)
Madani <i>et al.</i> (2025)	Madani et al.	Comprehensive Review (All Categories)

Table A.1: Methodological Classification of Cited Works

SUMMARY OF CLASSIFICATION

- **Model-Based:** Focuses on interpretability but lacks flexibility for unseen aging patterns.
- **Data-Driven:** Focuses on high accuracy using large datasets but lacks physical consistency.
- **Hybrid:** The target domain of this thesis, aiming to bridge the gap between the two above.

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